

GIS SPATIAL ANALYSIS

BY

DR. MEENAKSHI DEVI

EES, DEPARTMENT, IISER MOHALI

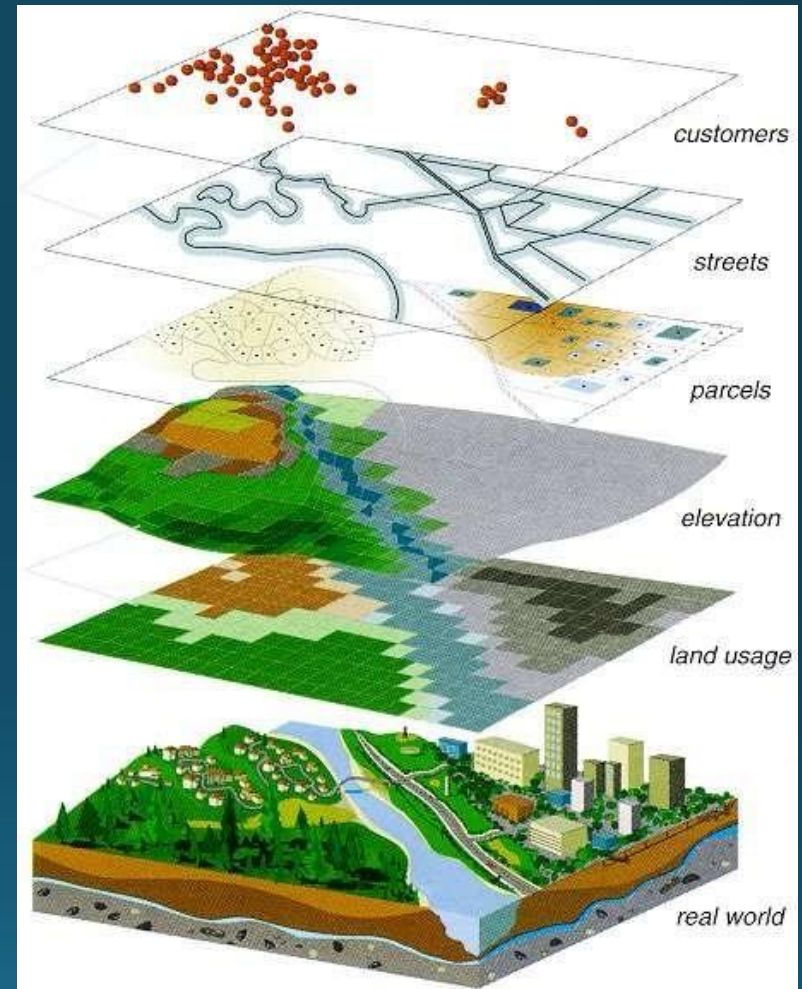
Spatial Data

- It can be most simply defined as information that describes the distribution of things upon the surface of the earth.
- In effect, it involves any information concerning the location, shape of, and relationships among, geographic features.



Spatial Analysis

- Manipulation of spatial data into various forms to be able to extract additional and meaningful information to understand the real-world.
- Can be highly technical/mathematical but can also be very simple and intuitive.



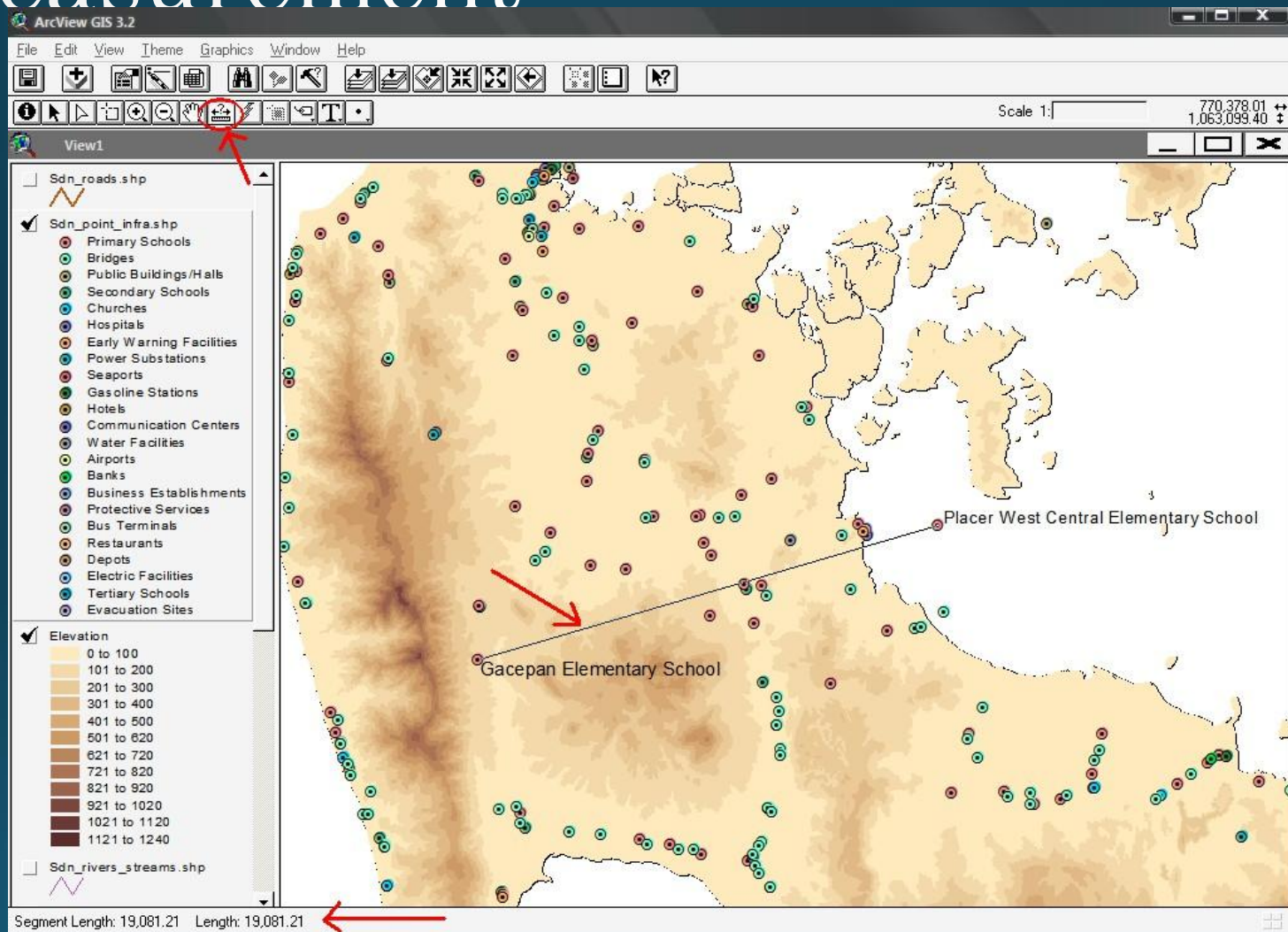
Common Examples of Spatial Data Analysis

- Measurements (lengths, perimeters, areas)
- Querying
- Reclassification
- Buffering and neighborhood functions
- Map overlay (vector overlay and raster overlay)
- Spatial interpolation
- Trend surface analysis
- Network analysis
- Suitability analysis

Measurement in GIS

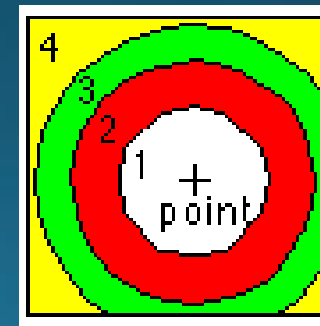
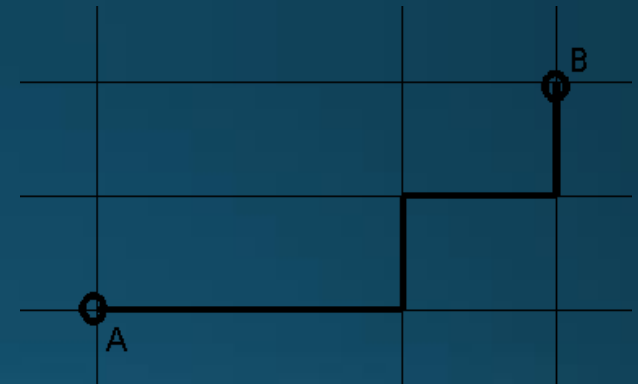
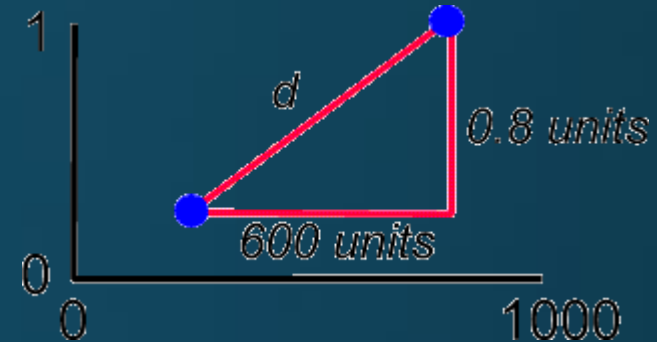
- Only an approximation of the true distance on the ground
 - Vector data are stored as line segments
 - even curved lines are stored as short line segments
 - Raster data are approximated using a grid cell representation
- Measurements can be stored as attributes in vector GIS

Distance Measurement



Distance Measurement of Raster Data

- Euclidean distance – shortest distance/path
- Manhattan distance – distance along raster cell sides from one point to another
- Proximity – concentric equidistant zones are established around the start point A, and the distance to B is taken to be the corresponding radius of the circle that intersects it



Perimeter Measurement of Raster Data

- Number of cell sides that make up the boundary of the feature is multiplied by the grid resolution

$$\text{Perimeter} = (\text{number of cell sides}) \times (\text{resolution})$$

Area Measurement of Raster Data

- Number of cells of a feature is multiplied by the known area of an individual grid cell

$$\text{Area} = (\text{number of cells}) \times (\text{area of individual cell})$$

Measurements of Vector Data

- Euclidean distance – obtained by using Pythagorean Theorem
- Perimeter - sum of the straight line lengths of the boundary
- Area - sum of areas of simple geometric shapes formed by subdividing the feature of interest

Querying

- Method of data retrieval
- Can be performed either on data that are part of the GIS database or on new data produced as a result of data analysis

Two types of query for GIS

1. Aspatial Query

- ❑ Questions about the attributes of features
- ❑ Example: “How many inter collages are there?”

2. Spatial Query

- ❑ Requires information about the location of a feature
- ❑ Example: “Where in Mohali are gas stations located?”

Combined Querying

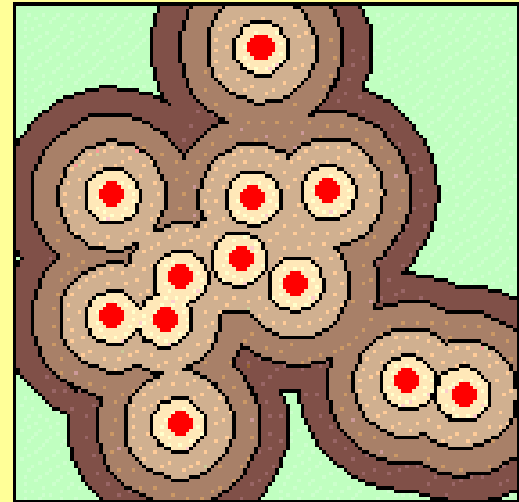
- Individual queries can be combined to identify features that satisfy two or more spatial and/or aspatial criteria
 - Example: “Where are the municipalities which have a population greater than 5000?”
- Usually uses Boolean operators (AND, OR, NOT)
Example: You have a GIS layer of land parcels with the following attributes:
LandUse (Forest, Agriculture, Urban)
Area (in hectares)
Q. Write a query to **select all parcels that are used for Agriculture AND have an area greater than 10 hectares.**
 - "LandUse" = 'Agriculture' AND "Area" > 10

Proximity Analysis (Buffering)

- Creation of a zone of interest around an entity
 - Point entity: circular buffer zone
 - Line entity: elongated buffer zone
 - Polygon entity: buffer zone has the same shape as original polygon, but larger

About buffers

Buffers are rings drawn around features at a specified distance from the features.

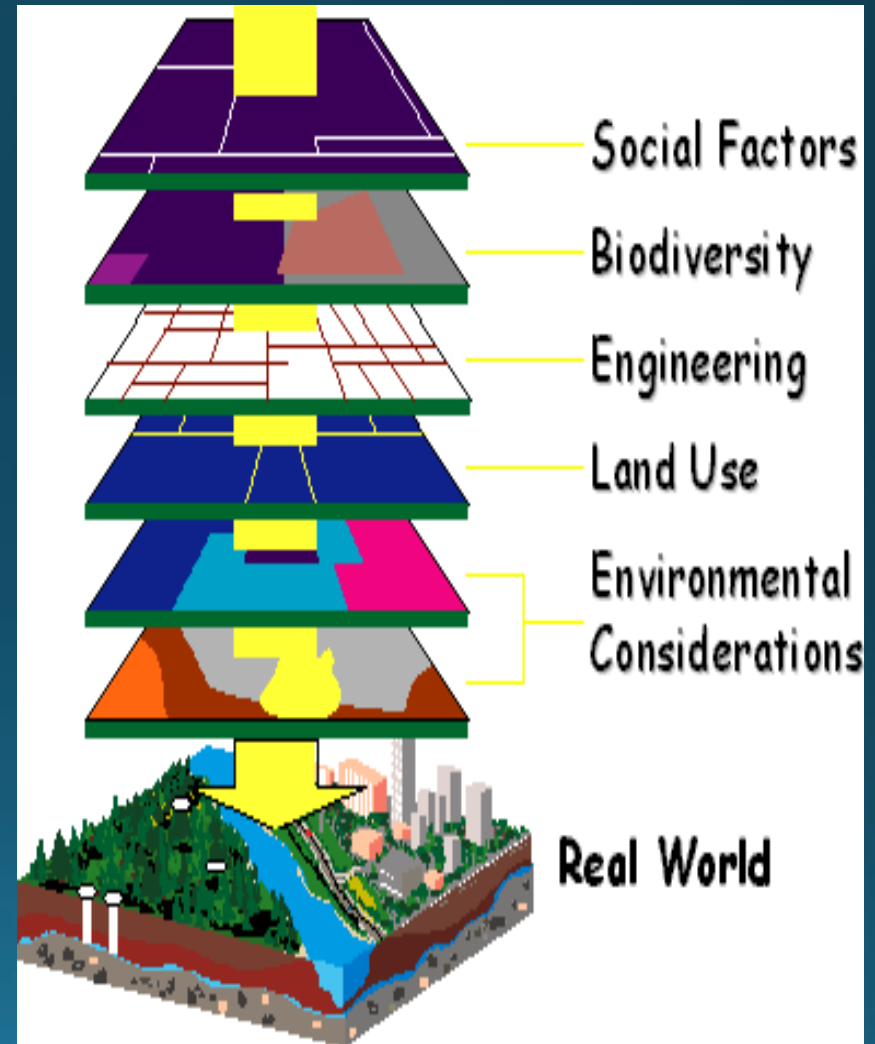


Sample Questions that Proximity Analysis Can Address

- How many houses lie within 100 m of this water main?
- What is the total number of patients within 10 km of this healthcare facility?
- What proportion of identified cases lies within 500 m of a suspected well (as source of infection)?

Topological Overlay

- Topology – relationships between linked elements
- Perhaps the most important or key GIS analysis function
- Integrating data from two or more sources to produce a new map layer
- More powerful when combined with other spatial analysis methods



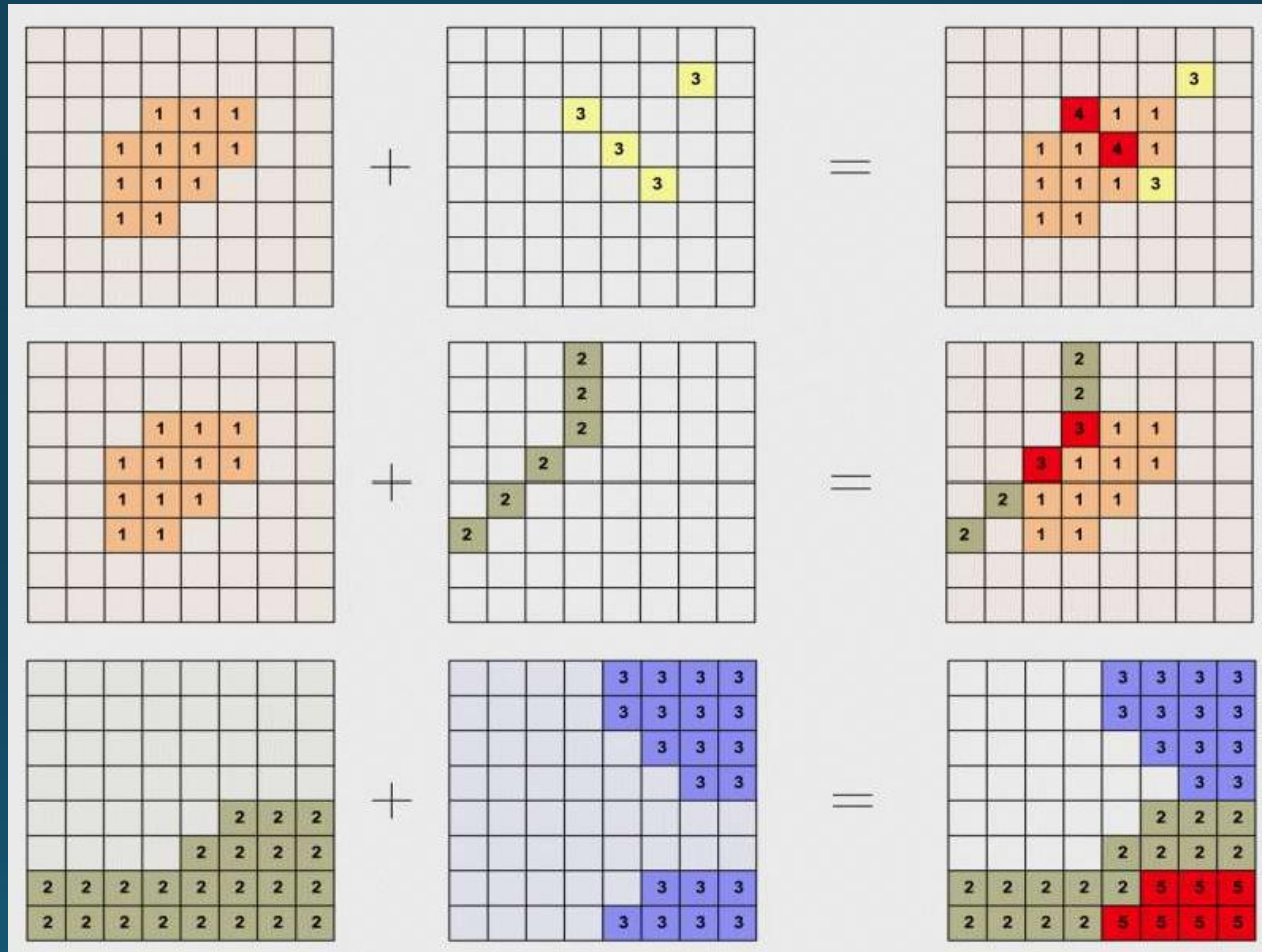
Raster Overlay

- Uses the idea of map algebra (“Mapemantics”)
- Mathematical operators are performed on corresponding cells from two or more layers to produce an output value
- Input layers may be added, subtracted, multiplied, or divided to produce output data
- Quick, straightforward, and efficient

Raster Overlay

- An input layer is usually reclassified by Boolean to make it more meaningful and easier to analyze
- It is important that the features on each layer are appropriately coded for easier analysis
- Lacks spatial accuracy for coarse cells/low resolution data

Raster Overlay



Vector Overlay

- Individual data layers to be overlaid have to be topologically correct (lines should meet at nodes and polygon boundaries are closed)
- Seldom used in isolation
 - commonly used with querying
- Time-consuming, complex, and computationally expensive

Vector Overlay

- Topology must be created for the resulting layer by calculating the intersections of lines and polygons from the input layers
 - ➔ time-consuming, especially for complex data
- Problem is the possible generation of sliver or 'weird' polygons from the overlay
 - ➔ results from overlaying two layers that contain the same spatial entities that are not exactly registered to one another

Types of Vector Overlay

1. Point-in-polygon
2. Line-in-polygon
3. Polygon-on-polygon

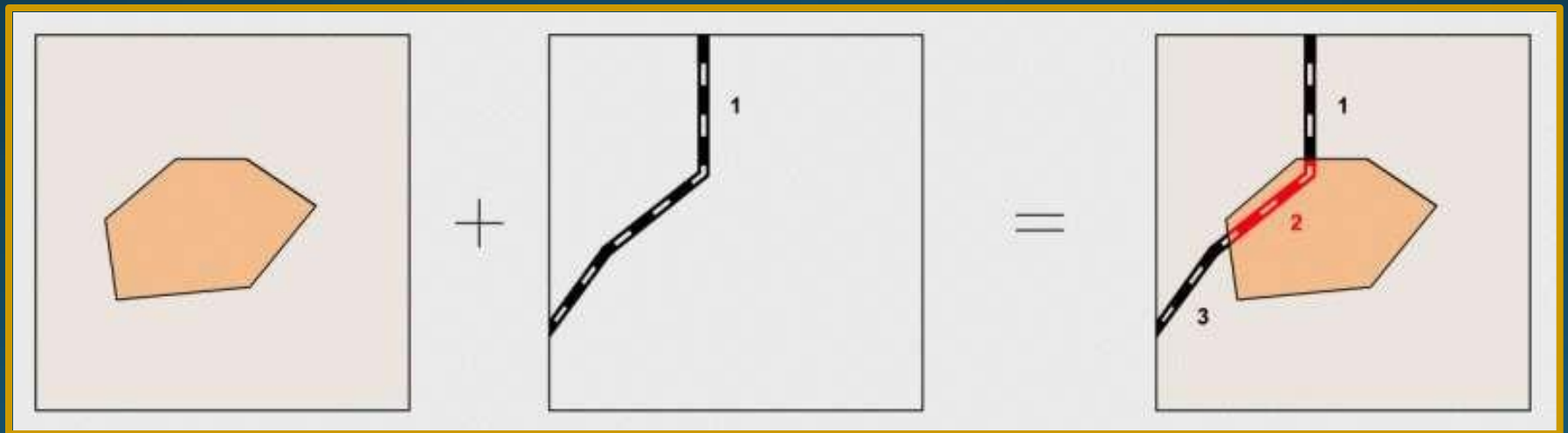
Point-in-Polygon Overlay

- used to find the polygon in which a point falls within, or find a point or points that fall within a certain polygon



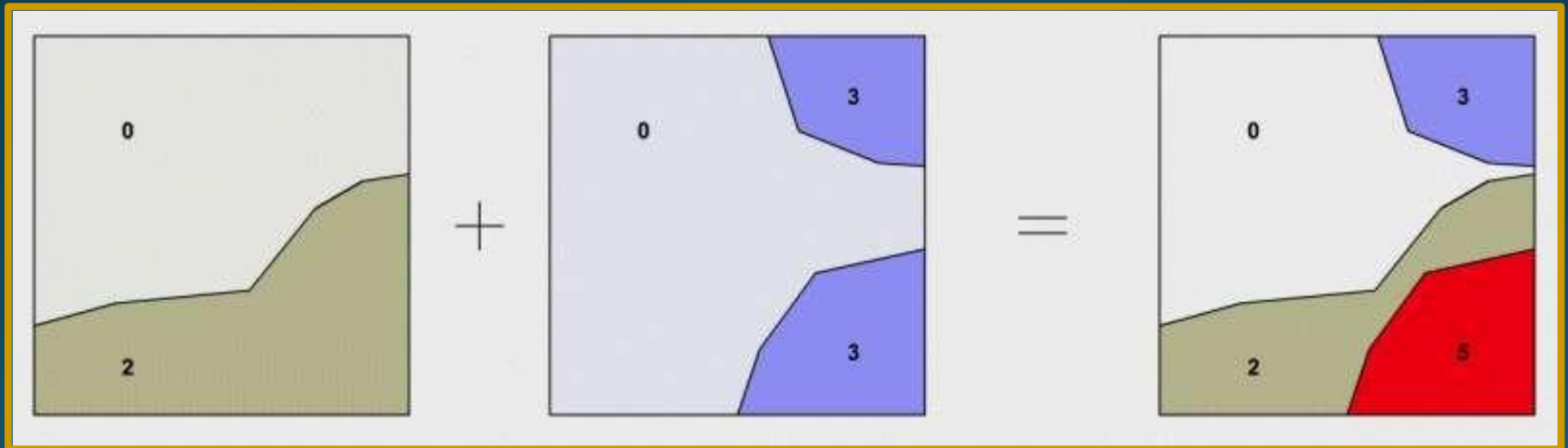
Line-in-Polygon Overlay

- used to find the polygon in which a line or lines fall within, or find a line or lines that fall within a certain polygon



Polygon-on-Polygon Overlay

- used to determine which polygons from two layers intersect or are within another polygon



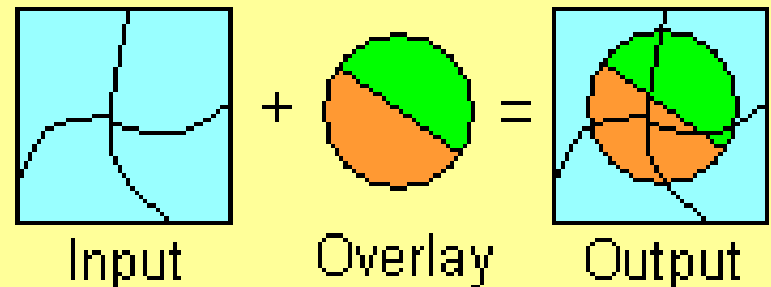
Types of Polygon-on-Polygon Overlay

- Union
- Clip/Cookie-cutting
- Intersect

Union

- Corresponds to the Boolean operator OR
- Output layer contains polygons from both input layers

This operation combines features of an input theme with the polygons from an overlay theme to produce an output theme that contains the attributes and full extent of both themes.



Clip/Cookie-cutting

- Corresponds to the Boolean operator NOT
- A polygon layer is used to cut out the portion of another polygon layer that falls within the first polygon

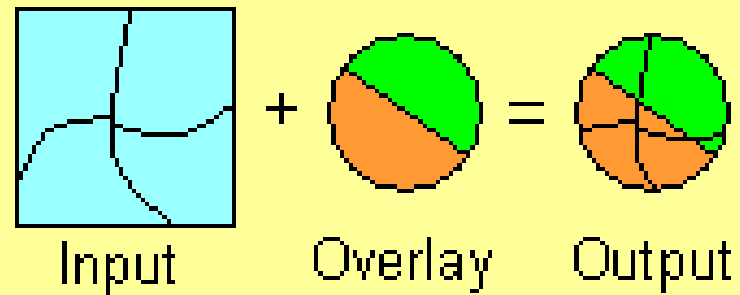
This operation uses a clip theme like a cookie cutter on your input theme. The input theme's attributes are not altered.



Intersect

- Corresponds to the Boolean operator AND
- Output is the polygon of intersection of two polygon layers

This operation cuts an input theme with the features from an overlay theme to produce an output theme with features that have attribute data from both themes.



Spatial Interpolation

- Procedure of estimating values at unsampled sites within an area with existing observations
- Used to fill the gaps between observed data points
- Interpolated data is only an approximation of the true value
- Commonly used to construct height contours

Spatial Interpolation Methods

1. Thiessen Polygons or Voronoi Polygons
2. Triangulated Irregular Network (TIN)
3. Spatial Moving Average

Thiessen Polygons

- Used to establish area territories for a set of points
- Examples: transformation of point climate stations to watersheds; construction of areas of influence around population centers

Steps in creating Thiessen Polygons

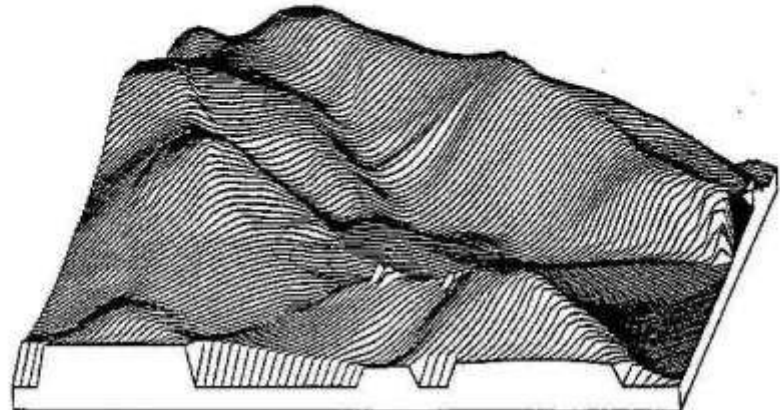
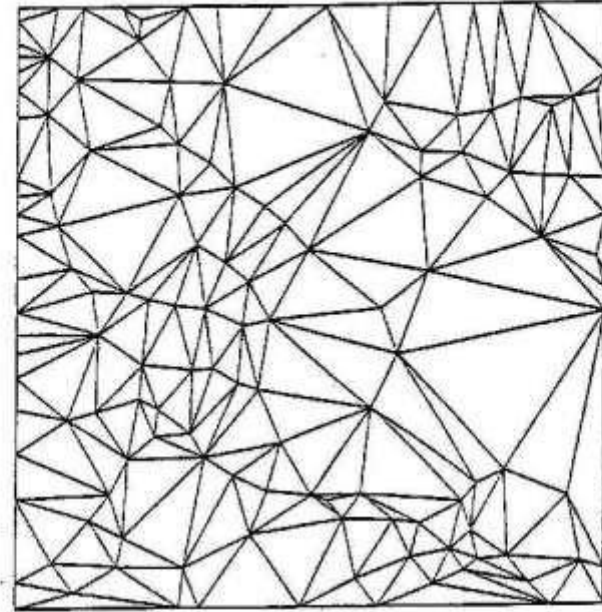
- Draw lines joining nearest neighboring points
- Draw perpendicular bisectors of these lines
- Use the bisectors as edges of the polygons that make up the surface



Triangulated Irregular Network

- An elegant way of constructing a surface from a set of irregularly spaced data points
- Most often used to generate DEMs
- Exact interpolation based on local data points
- Adjacent data points are connected by lines to form a network of irregular triangles

Triangulated Irregular Network

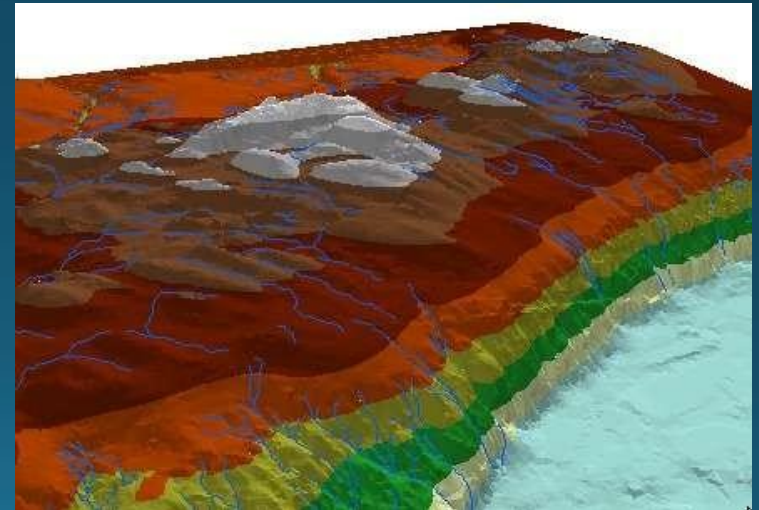
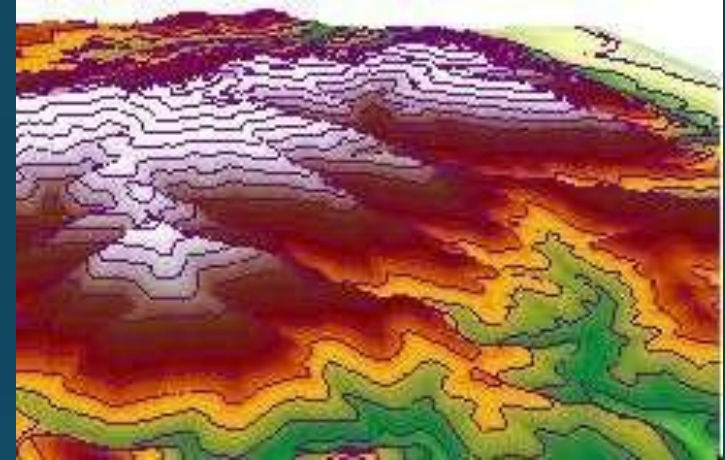
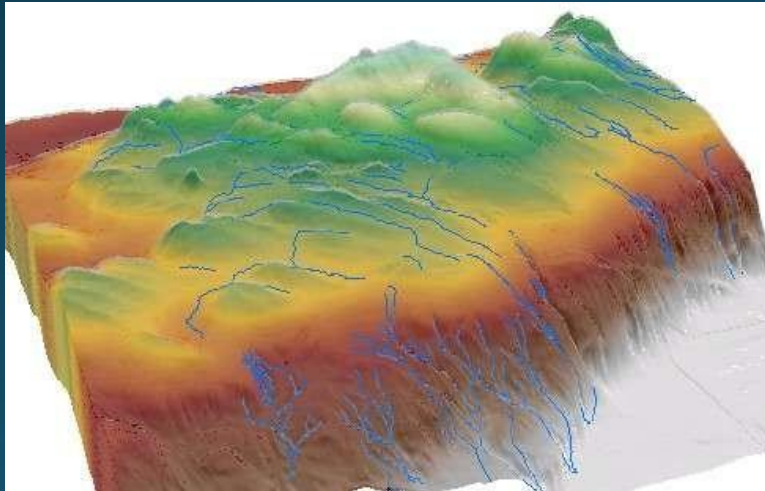


Spatial Moving Averages

- An approximate interpolation $\square$ does not honor the values of known data points (recalculates them)
- Used when values of known data points are not exact/erroneous
- Examples: interpolation of census data, questionnaire responses, soil PH, rainfall infiltration rates

Surface Analysis

- Slope and aspect calculation
- Visibility analysis



Visibility Analysis

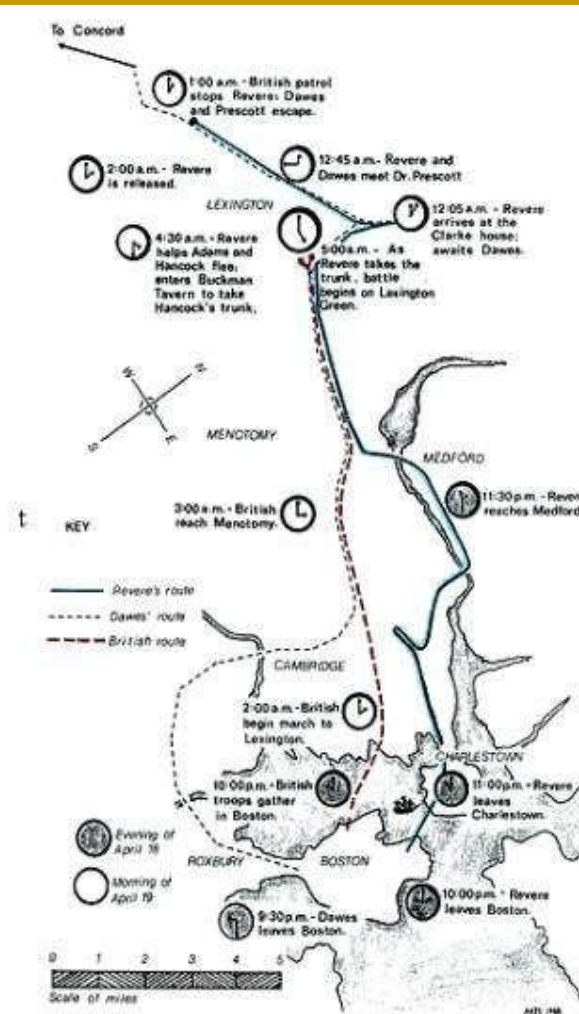
- Identification of areas of terrain that can be seen from a particular point on the terrain surface
- Applications:
 - Locating towers for geodetic surveying
 - Locating radio transmitters for maximum coverage

Network analysis

- Involves linear features or network of linear features that are topologically structured
 - roads, rivers, pipelines, cables, etc.
- Examples:
 - shortest path
 - time of travel
 - route tracing

Network Analysis Example

Directions	Distance
1: Start out going East on HENLEY ST towards WARREN ST.	0.1 miles (0.1 km)
2: Turn RIGHT onto WARREN ST.	0.0 miles (0.1 km)
3: Turn RIGHT onto CHELSEA ST.	0.0 miles (0.1 km)
4: CHELSEA ST becomes CHELSEA ST/CITY SQ.	0.1 miles (0.1 km)
5: Turn RIGHT onto CITY SQ/NEW RUTHERFORD AVE/SR-99 N.	0.0 miles (0.1 km)
6: Stay straight to go onto NEW RUTHERFORD AVE/SR-99 N.	0.2 miles (0.3 km)
7: Turn SLIGHT LEFT onto SR-99 N.	0.4 miles (0.6 km)
8: Turn SLIGHT LEFT onto SR-99 N/RUTHERFORD AVE.	0.1 miles (0.1 km)
9: Turn SLIGHT LEFT onto SR-99 N.	0.3 miles (0.4 km)
10: Turn SLIGHT LEFT onto SULLIVAN SQUARE OPAS.	0.4 miles (0.7 km)
11: Turn SLIGHT LEFT onto MYSTIC AVE.	0.7 miles (1.1 km)
12: MYSTIC AVE becomes MYSTIC AVE/SR-38 N.	1.2 miles (2.0 km)
13: Turn LEFT onto HARVARD ST.	0.6 miles (1.0 km)
14: HARVARD ST becomes WARNER ST.	0.2 miles (0.3 km)
15: Turn RIGHT onto POWDER HOUSE SQ.	0.1 miles (0.1 km)
16: Turn RIGHT onto BROADWAY.	1.0 miles (1.6 km)
17: Turn LEFT onto ALEWIFE BROOK PKWY/SR-16.	0.4 miles (0.7 km)
18: ALEWIFE BROOK PKWY/SR-16 becomes ALEWIFE BROOK PKWY/SR-16/US-3.	0.4 miles (0.7 km)
19: Take CONCORD TURNPIKE/SR-2 W.	4.7 miles (7.6 km)
20: Take the WALTHAM ST. exit, exit number 54B, towards LEXINGTON.	0.2 miles (0.3 km)
21: Merge onto WALTHAM ST.	1.9 miles (3.0 km)
22: Turn RIGHT onto MASSACHUSETTS AVE/MASS AVE/SR-225.	0.0 miles (0.0 km)
Total Distance:	12.9 miles (20.8 km)
Estimated Time:	24 minutes





Thank you!



Reclassification

- An important variation of querying in GIS
- Can be used in place of querying for raster GIS
- Uses reclassification rules to reclassify or assign new values to cells
- Results to a new image
- May be Boolean or Weighted

Boolean Reclassification

- Produces a two-coded image from a complex image
- Original image is reclassified to an image with only 0 or 1 as cell values
- Example, if the production forests are of interest, all cells representing forestry can be assigned a value of 1, and all other cells a value of 0.

Reclassification by Weighting

- A different weight is assigned to different feature types or classes, based on the purpose of the reclassification
- Higher weights can be assigned to priority classes while lower weights to those of lower priority
- Example, if the purpose is for forest conservation, a weight of 5 can be assigned to forestry and lower weights to other features

Reclassification Example

Land use	Original value	New value: Boolean	New value: Weighted
Forestry	10	1	4
Water	11	0	2
Settlement	12	0	1
Agricultural Land	13	0	3



Digital Terrain Analysis

BY

DR. MEENAKSHI DEVI

EES DEPARTMENT, IISER MOHALI

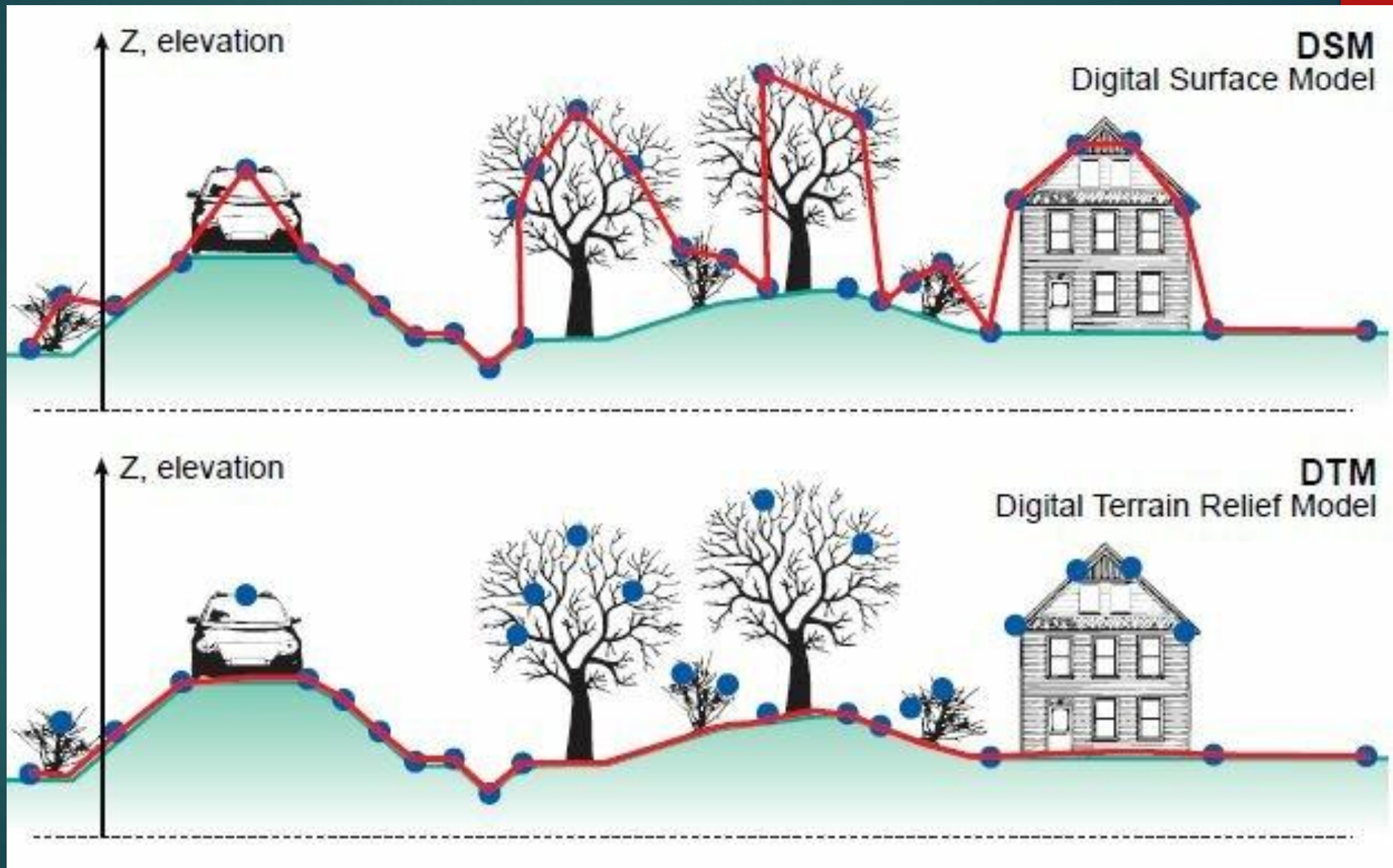
1. Introduction

- **Digital Terrain Analysis (DTA):**
The study and quantitative evaluation of the Earth's surface topography using digital data, usually derived from **Digital Elevation Models (DEMs)**.
- Provides a framework for understanding **landforms, surface processes, and spatial patterns** of terrain.
- Widely used in **geomorphology, hydrology, natural hazard studies, and environmental planning**.

Digital Elevation Model (DEM)



DSM vs DTM



2. Data Sources for DTA

- ▶ **Digital Elevation Models (DEM):** Grid-based representation of terrain heights.
 - ▶ Examples: SRTM, ASTER GDEM, LiDAR DEMs, CartoDEM (India).
- ▶ **Digital Surface Models (DSM):** Represent surface including vegetation and buildings.
- ▶ **Remote Sensing & Photogrammetry:** Satellite stereo pairs, aerial photography, UAV-based data.
- ▶ **Topographic Survey Data:** Ground-based surveying and GPS.

3. Fundamental Terrain Parameters

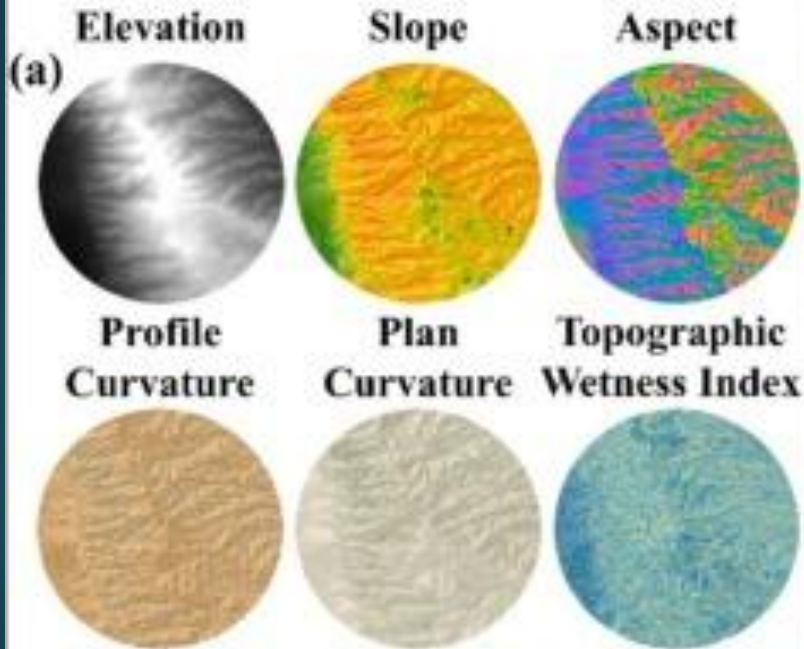
- ▶ Derived directly from DEMs:
- ▶ **Elevation** – Height above a reference (sea level).
- ▶ **Slope** – Steepness/gradient of terrain.
- ▶ **Aspect** – Direction of slope (N, NE, S, etc.).
- ▶ **Curvature** – Convexity or concavity of surface (plan/profile curvature).
- ▶ **Topographic Roughness/Texture** – Variability in elevation.

4. Secondary Terrain Parameters

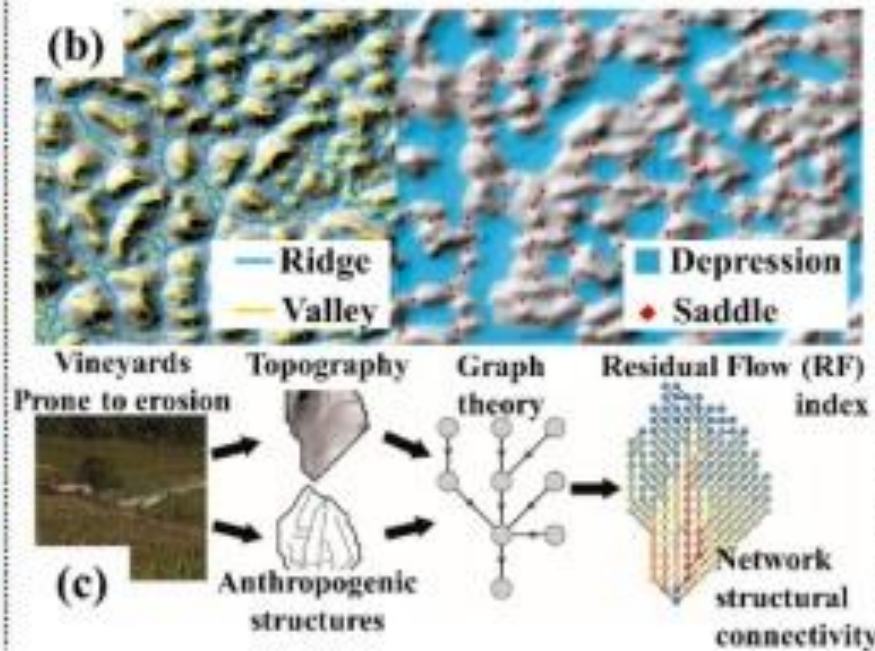
- ▶ Derived from combinations of fundamental parameters:
- ▶ **Topographic Wetness Index (TWI):** Indicates soil moisture potential.
- ▶ **Stream Power Index (SPI):** Reflects erosive power of water.
- ▶ **Landform Classification:** Hilltops, ridges, valleys, plains, depressions.
- ▶ **Morphometric Parameters:** Basin shape, hypsometric curve, drainage density, relief ratio.

Digital Terrain Analysis

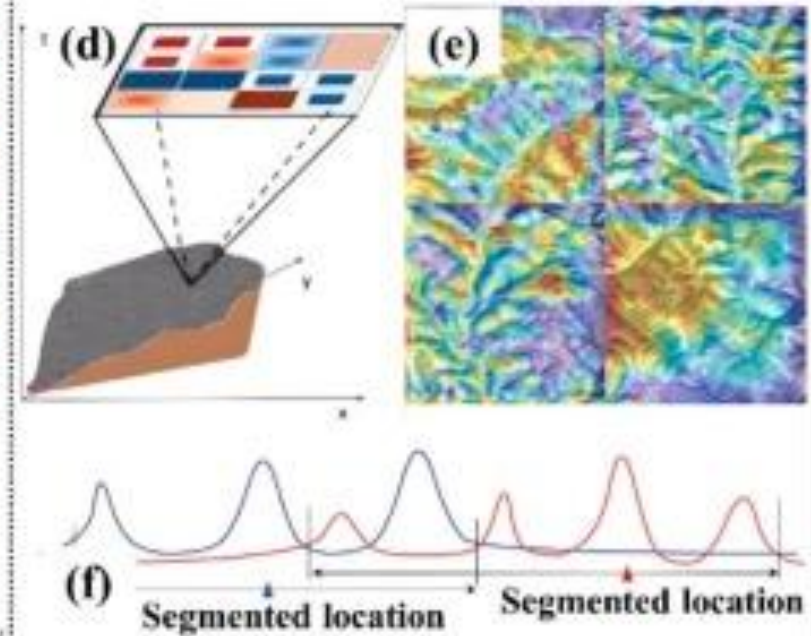
Topographic Attributes



Topographic Features/Objects



Topographic Information



Slope

- ▶ **What it is:** The steepness of terrain, usually expressed in degrees or percent rise.
- ▶ **Formula:**
- ▶
$$\text{Slope} = \tan^{-1} \left(\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \right)$$
- ▶ where $\frac{\partial z}{\partial x}$ (east-west) and $\frac{\partial z}{\partial y}$ (north-south) are changes in elevation along X and Y.
- ▶ **In GIS:**
 - ▶ ArcGIS: *Spatial Analyst* → *Surface* → *Slope*
 - ▶ QGIS: *Raster* → *Terrain Analysis* → *Slope*
 - ▶ Output: New raster showing slope value for each cell.

► **Example: Slope Calculation in a 3×3 Grid**

- Suppose we have a **3×3 DEM window** (elevation values in meters).
The center cell (Z_5) is where we want to calculate slope:

Z_1	Z_2	Z_3
120	122	121
118	119	117
116	115	114

Here:

- Cell size = **30 m** (typical SRTM resolution).
- We calculate slope at **$Z_5 = 119$** (the center cell).

Using Horn's method (commonly used in GIS software):

$$\begin{aligned} dz/dx &= \frac{(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)}{8d} \\ dz/dy &= \frac{(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)}{8d} \end{aligned}$$

Flow Direction

- ▶ Flow direction determines the path water would take as it flows across a DEM surface.
- ▶ It's a cell-based operation: for each grid cell, the algorithm finds which neighboring cell has the steepest downslope descent.
- ▶ Output = raster where each cell stores a code indicating the direction of flow.
- ▶ Formula:
- ▶ $\text{slope} = (\text{zcenter} - \text{zneighbor}) / \text{Distance}$
- ▶ Example
- ▶ DEM (3×3 grid, cell size = 30 m):

100	95	90
105	80	70
110	120	130

- ▶ Calculate flow direction for cell=80 ? ?
- ▶ Given cell size 30m

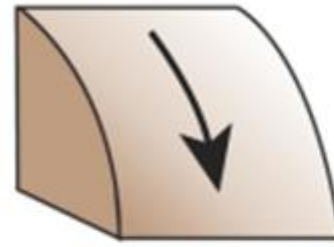
Aspect

- ▶ **What it is:** Direction of slope (0° = North, 90° = East, etc.).
- ▶ **Formula:**
- ▶ $Aspect = \tan^{-1} \left(\frac{\partial z / \partial y}{-\partial z / \partial x} \right)$
- ▶ **In GIS:**
 - ▶ ArcGIS: *Spatial Analyst* → *Surface* → *Aspect*
 - ▶ QGIS: *Raster* → *Terrain Analysis* → *Aspect*
 - ▶ Output: Raster with aspect values (flat areas marked as “-1” or “no aspect”).

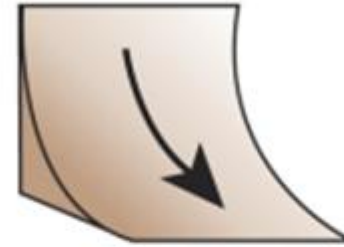
Curvature

- ▶ It indicates whether a slope is concave (like a bowl, retaining water) or convex (like a bulge, causing water runoff).
- ▶ **Types:**
 - ▶ **Profile curvature** → affects flow acceleration/deceleration.
 - ▶ **Plan curvature** → affects flow convergence/divergence.
- ▶ **In GIS:**
 - ▶ ArcGIS: *Spatial Analyst* → *Surface* → *Curvature*
 - ▶ QGIS/SAGA: *Raster Terrain Analysis* → *Curvature*

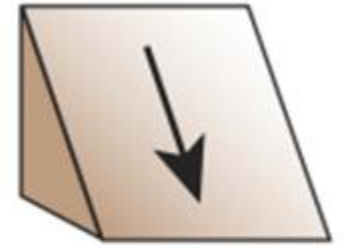
Profile curvature



-

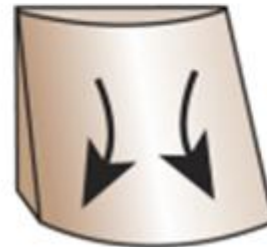


+



0

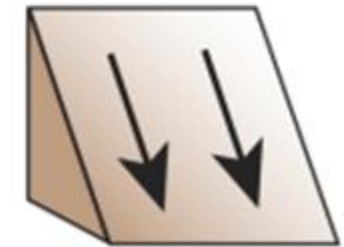
Plan curvature



+



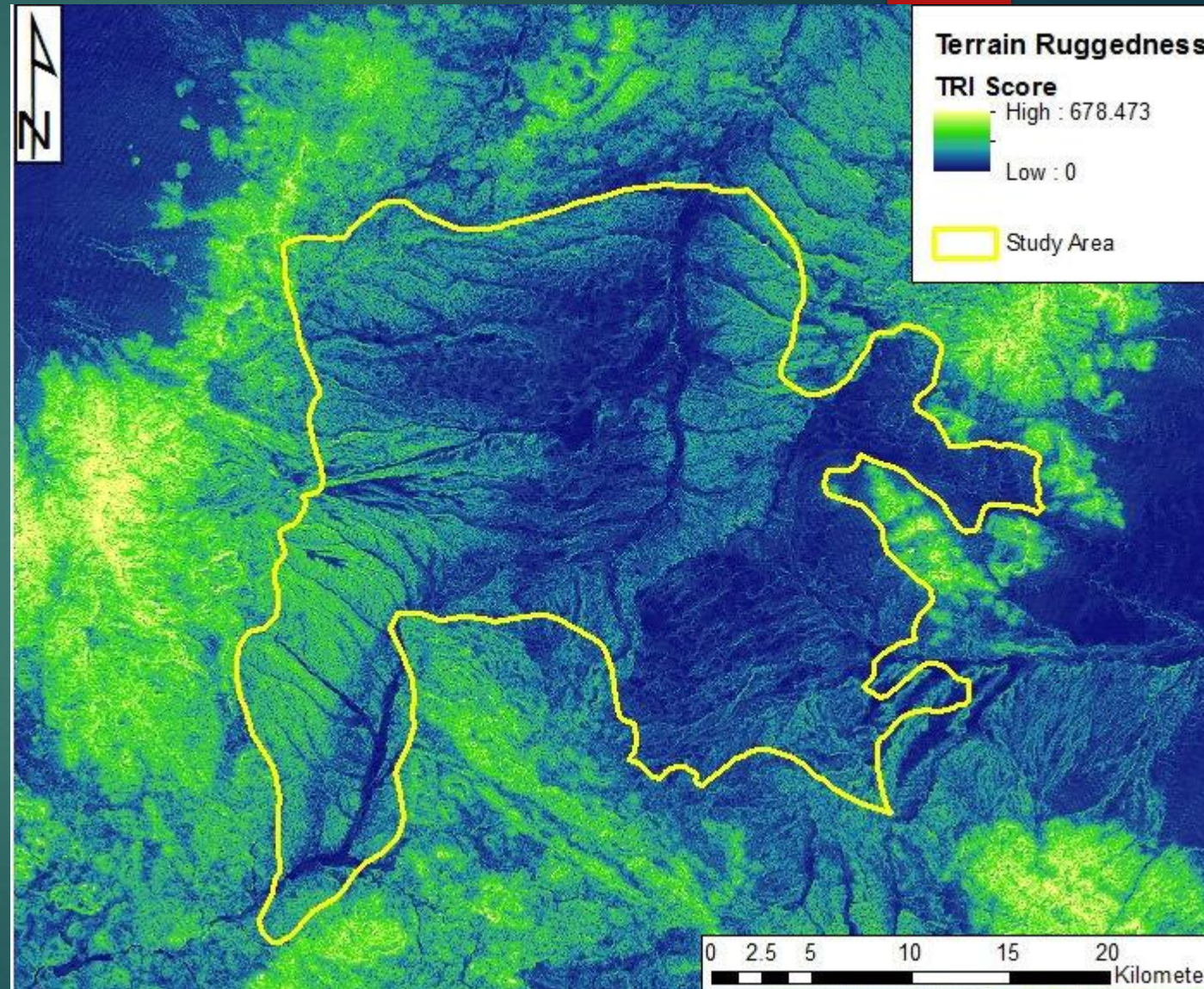
-



0

Roughness / Terrain Ruggedness

- ▶ Quantifies the amount of elevation change between a cell and its neighbors in a Digital Elevation Model (DEM).
- ▶ **Methods:**
 - ▶ Range of elevation values in a neighborhood.
 - ▶ Standard deviation of slope/elevation.
- ▶ **In GIS:**
 - ▶ QGIS: *Ruggedness Index Plugin*
 - ▶ ArcGIS: *Surface Roughness* or *Terrain Ruggedness Index (TRI)* tools.



Topographic Wetness Index (TWI)

- ▶ **What it is:** Indicator of soil moisture/water accumulation potential.
- ▶ **Formula:**
- ▶ $TWI = \ln \left(\frac{a}{\tan \beta} \right)$
- ▶ where:
 - ▶ a = upslope contributing area (catchment area per unit contour length)
 - ▶ β = slope angle (in radians)
- ▶ **In GIS:**
 - ▶ Extract flow accumulation (from DEM).
 - ▶ Calculate slope raster.
 - ▶ Apply formula with raster calculator.

Stream Power Index (SPI)

- ▶ **What it is:** Represents erosive power of flowing water.
- ▶ **Formula:**
- ▶ $SPI = a \cdot \tan \beta$
 - ▶ a = flow accumulation
 - ▶ β = slope angle
- ▶ **In GIS:**
 - ▶ Same input rasters as TWI.
 - ▶ Multiply flow accumulation raster with slope raster.
- ▶ **Example:**
 - ▶ If $a = 500\text{m}^2$, slope = 10°
- ▶ → Higher SPI = more erosive flow (steep valley).

Morphometric Parameters (for basins/landforms)

- ▶ Derived from DEM using watershed analysis:

- ▶ **Drainage Density (Dd):**

- ▶ $D_d = \frac{L}{A}$

- ▶ L = total length of streams

- ▶ A = basin area

- ▶ **Relief Ratio (Rr):**

- ▶ $R_r = \frac{H}{L_b}$

- ▶ $H = h_{max} - h_{min}$

- ▶ L_b = basin length

Hydrologic functions

▶ **Fill (Sink Removal)**

- ▶ DEMs often have artificial pits or depressions due to errors.
- ▶ *Fill* operation removes these sinks to create a hydrologically correct surface.
- ▶ Ensures continuous flow across DEM.

▶ **Flow Direction**

- ▶ Determines the direction of water flow from each cell.
- ▶ Uses methods like:
 - ▶ **D8** (steepest descent to one of 8 neighbors).
 - ▶ **D ∞** (flow distributed between downslope neighbors).
- ▶ Output: Raster showing flow direction codes.

Flow Accumulation

- Calculates how many upstream cells contribute flow into each cell.
- High values = valleys/stream channels.
- Low values = ridges/divides.
- Used for stream network extraction.

Stream Definition and Ordering

- Converts flow accumulation raster into a **stream network** using threshold values.
- Stream ordering methods:
 - **Strahler Order** (streams of same order join → order increases).
 - **Shreve Order** (order = sum of upstream tributaries).

Watershed Delineation

- ▶ Identifies the area draining to a particular point (pour point).
- ▶ Based on flow direction and accumulation.
- ▶ Useful for catchment-scale hydrological studies.

Flow Length

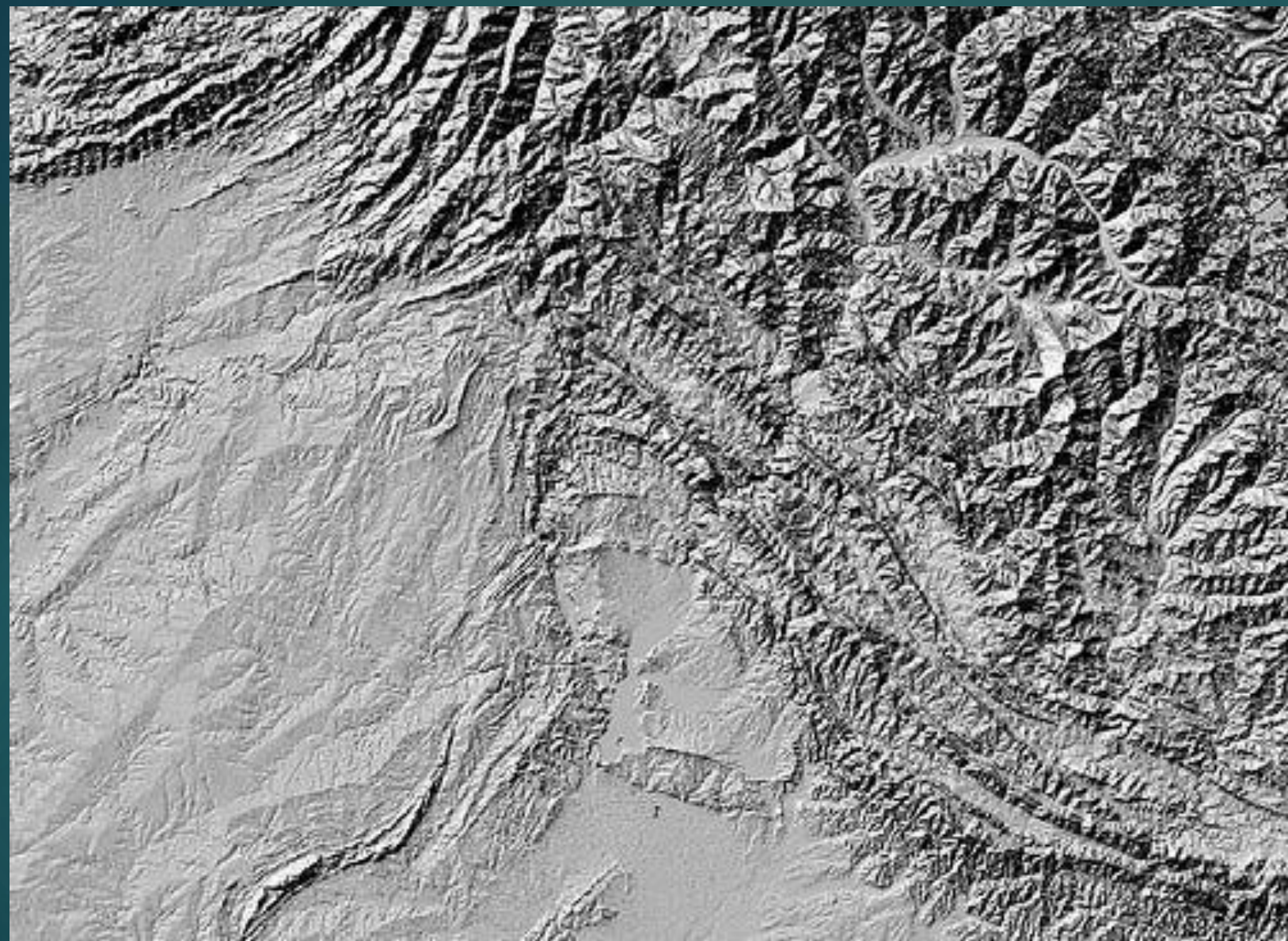
- Calculates distance water will travel:
 - **Downslope flow length** → from cell to outlet.
 - **Upslope flow length** → from cell to ridgeline.
- Useful for erosion modeling.

Stream and Basin Characteristics

- ▶ Extraction of morphometric properties:
 - ▶ Stream length, density, bifurcation ratio.
 - ▶ Basin area, relief, shape factor.
- ▶ Used in hydrological modeling and watershed management.

Shaded relief map

- ▶ A **shaded relief map** (also called a **hillshade map**) is a cartographic technique used to give a **3D-like appearance** to terrain.
- ▶ It simulates how the terrain would look if illuminated by a hypothetical light source (usually the sun).
- ▶ Widely used in **Digital Terrain Analysis** to visualize landforms and enhance the interpretability of DEMs.
- ▶ **Principle**
- ▶ Based on **illumination and shadow effects** on terrain surfaces.
- ▶ The appearance of each DEM cell depends on:
 - ▶ **Slope** – steepness of terrain.
 - ▶ **Aspect** – direction the slope faces.
 - ▶ **Sun's position** – defined by **azimuth** (horizontal angle of sunlight) and **altitude** (height of sun above horizon).



Profile Plots

- A **profile plot** (also called a cross-section or elevation profile) is a **2D graph showing changes in elevation along a line or path across terrain**.
- It is one of the simplest but most powerful visualization tools in Digital Terrain Analysis.
- ▶ Helps in understanding the **vertical dimension of terrain** that is not easily visible in 2D maps.

2. How Profile Plots are Created

- ▶ **Input Data:** Digital Elevation Model (DEM).
- ▶ **Draw a Line/Transect:** User specifies a line across the DEM (e.g., from one hilltop to a valley).
- ▶ **Interpolation:** Elevation values are extracted along the line from DEM cells.
- ▶ **Graph Plotting:** Elevation (Y-axis) vs. Distance (X-axis) is plotted.

3. Types of Profile Plots

- ▶ **Simple Profile:** Straight line between two points (e.g., ridge to valley).
- ▶ **Longitudinal Profile:** Along a river/stream course (used in fluvial geomorphology).
- ▶ **Transverse Profile:** Across a valley, ridge, or dune (used in geomorphic cross-sections).

5. Applications of DTA

▶ (a) Geomorphology

- ▶ Mapping landforms (valleys, ridges, terraces, floodplains).
- ▶ Quantitative analysis of slope evolution and landscape processes.

▶ (b) Hydrology

- ▶ Watershed delineation.
- ▶ Stream network extraction.
- ▶ Flood risk and erosion hazard assessment.

(c) Soil and Agriculture

Identifying soil moisture zones.
Locating potential sites for agriculture and irrigation.

(d) Environmental & Hazard Studies

Landslide susceptibility mapping.
Volcanic hazard and lava flow modeling.
Coastal and desert dune analysis.

(e) Urban and Engineering Applications

Site suitability analysis for roads, dams, and settlements.
Terrain stability and slope analysis for construction projects.



Geospatial Modelling

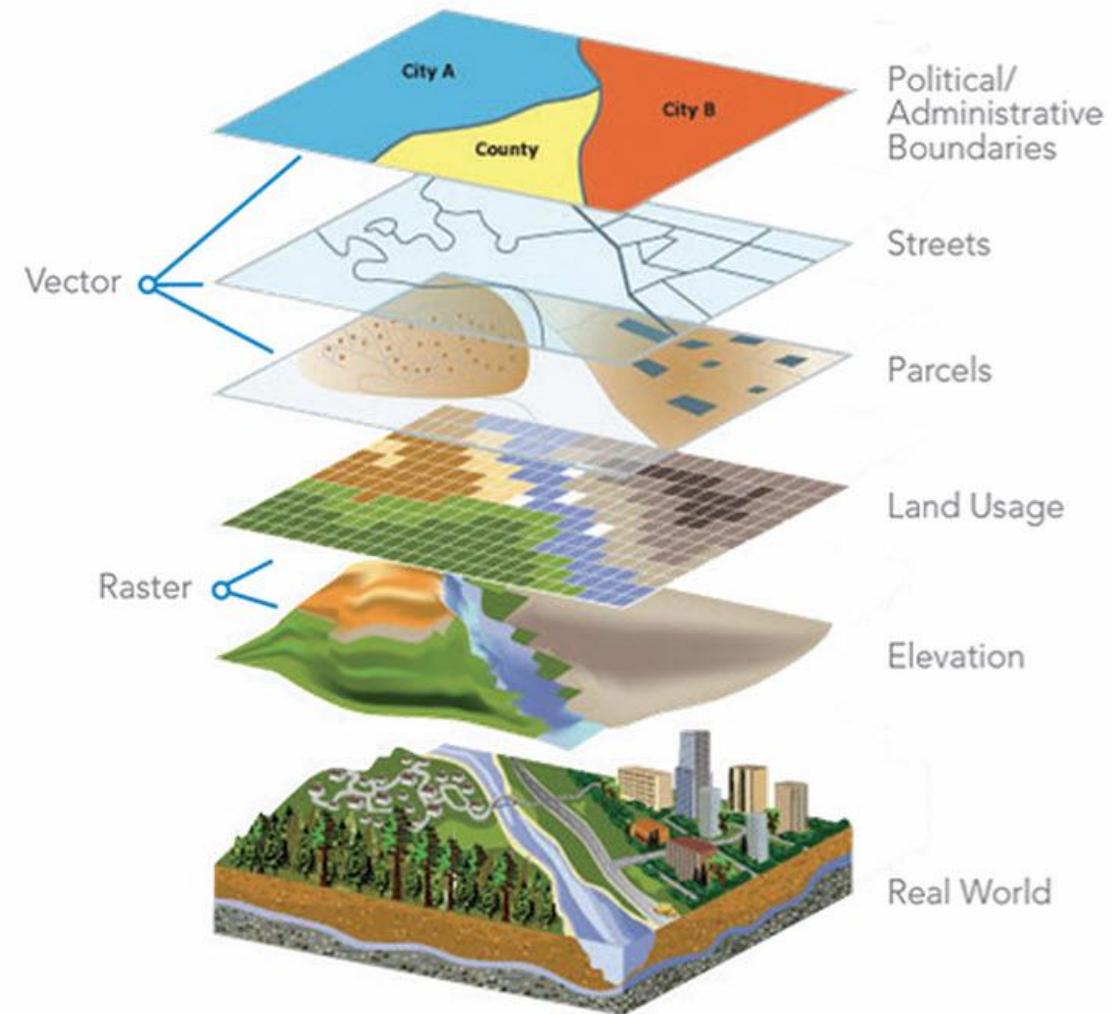
BY

DR. MEENAKSHI DEVI

EES DEPARTMENT, IISER MOHALI

Introduction: What is Spatial Modeling?

- ▶ Spatial modeling is the analytical process of using GIS to apply a set of rules and procedures to spatial data to answer a question, solve a problem, or simulate a process.
- ▶ **Analogy:** Think of it as a "recipe." You have **ingredients** (input data) and a set of **instructions** (the model's rules and procedures). Following the recipe gives you a final **dish** (the output or result).
- ▶ **Formal Definition:** It is a simplified, quantitative representation of the real world, constructed to illustrate, predict, or simulate spatial phenomena under specific assumptions.



Why is it Important?

- ▶ **Goes beyond mapping:** It moves from "where" things are to "why" they are there, "what if" scenarios, and "what is best."
- ▶ **Decision Support:** Provides a scientific basis for complex decisions in urban planning, environmental management, public health, and business.
- ▶ **Prediction & Forecasting:** Allows us to anticipate future events, like urban sprawl, flood risk, or disease outbreaks.
- ▶ **Efficiency:** Automates complex, multi-step analyses that would be tedious or impossible to do manually.

Static Models

Definition:

- A *static model* represents a system at a **single point in time**.
- It combines **multiple inputs** into a **single output**.
- There are **no time steps** or **loops** involved.
- **Example – Universal Soil Loss Equation (USLE):**
- **Purpose:** Predicts soil loss at a point based on five input variables.
- **Equation:**

$$A=R \times K \times LS \times C \times P$$

A – Predicted erosion rate

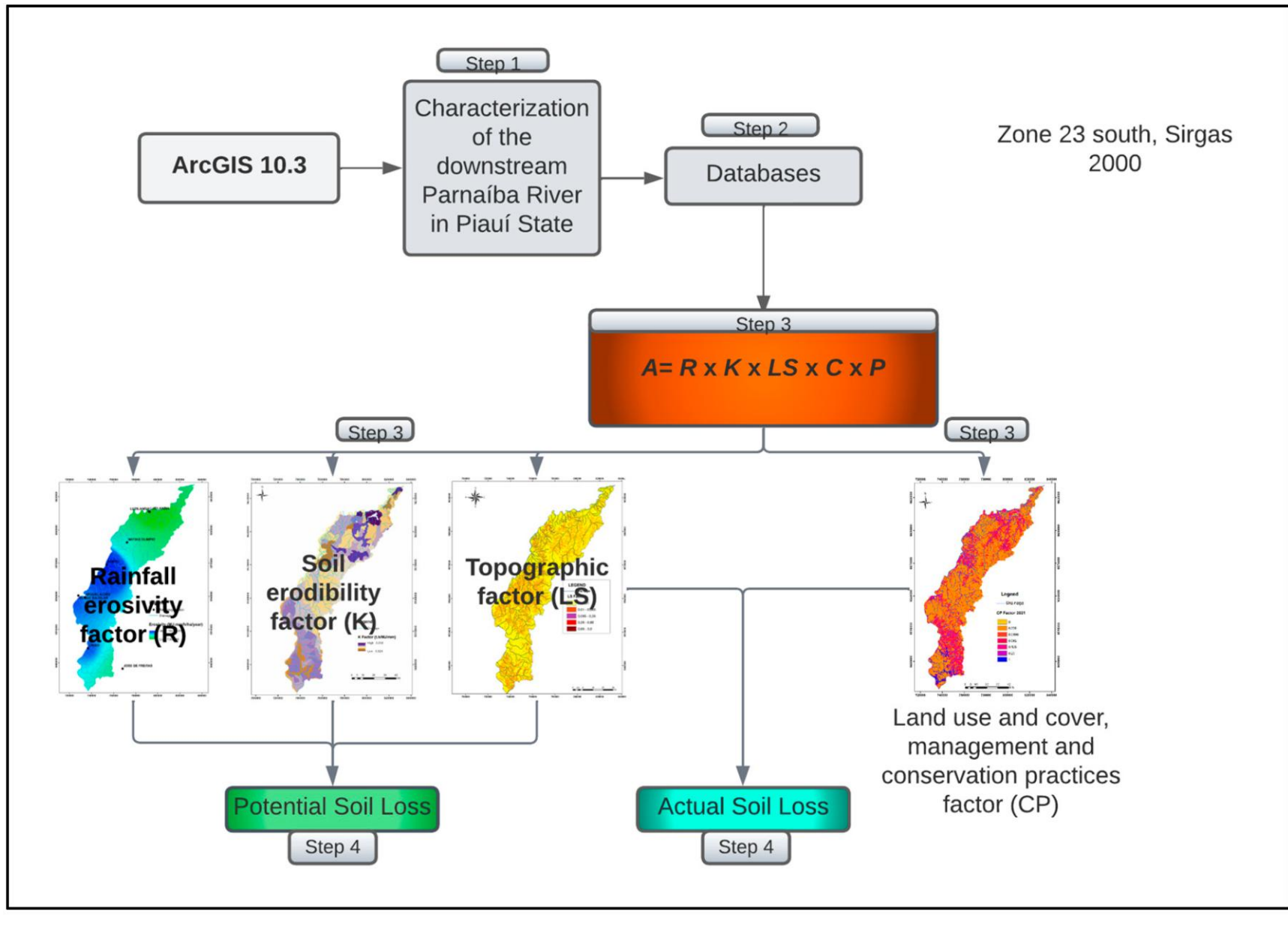
R – Rainfall and Runoff Factor

K – Soil Erodibility Factor

LS – Slope Length Gradient Factor

C – Crop/Vegetation and Management Factor

P – Support Practice Factor



Broader Use of Static Models:

- Common in **environmental modeling** to combine various input layers into an output index or indicator.
- **Examples of Other Static Models:**
- **DRASTIC Model:**
 - Calculates an **index of groundwater vulnerability**.
 - Uses input layers with **weighted parameters**.
- **Groundwater Protection Model (Karst Environment):**
 - Combines several inputs in areas underlain by **soluble limestone** with **rapid groundwater flow** through caves.

GIS
Database



D - Depth to water table

R - Net recharge

A - Aquifer media

S - Soil media

T - Topography

I - Impact to vadose zone

C - Hydraulic conductivity

Individual vulnerability maps

Aquifer vulnerability map

Types of Spatial Models

1 Deterministic Models

- Based on known physical laws and equations.
- Example: Runoff estimation using the SCS-CN method in hydrological modelling.

2 Stochastic Models

- Incorporate randomness and probability.
- Example: Predicting landslide susceptibility using logistic regression.

3 Empirical Models

- Based on statistical relationships derived from observed data.
- Example: Land surface temperature modelling using regression of NDVI and albedo.

Cellular Models or Agent based Model

Definition:

- Represent the **Earth's surface as a raster** (a grid of cells).
- Each **cell** has a **number of possible states** that can **change through time**.
- Changes occur due to **transition rules** applied to each cell and its **neighborhood**.
- **Key Characteristics:**
 - Rules determine outcomes based on:
 - The **current state** of the cell.
 - The **states of neighboring cells**.
 - The **values of cell attributes** (e.g., elevation, land use, etc.).
 - The model operates through **iterations**—each step updates the states according to rules.

Applications of Cellular Models

- **Urban Growth Simulation:**

- Widely used to **simulate and predict urban development.**

- Factors influencing land development include:

 - **LULC**

 - **Access to transportation routes**

 - **Zoning or conservation status**

 - **Proximity to existing development** (most influential)

- **Rule Modification:**

- Similar to Conway's Game of Life:

 - The **more developed neighboring cells**, the **greater the chance** a cell transitions from **undeveloped → developed.**

Model Calibration and Validation

Calibration:

- Essential to ensure **rules and parameters** are realistic.
- **Brute-force calibration:**
 - Ran the model from a past date using many rule combinations.
 - Compared simulated results with **actual historical development**.
- This process is **time-consuming** but builds **confidence** in model accuracy.

Validation:

- Confirms whether the model's **predictions match real-world data**.
- A **critical step** in all modeling processes.

• **Summary Statement:**

- *A model must be calibrated against real data to determine valid parameters and ensure accurate simulation results.*

Script-Based Modeling

- Scripts are **functionally equivalent** to visual models but **less intuitive**.
- They require more technical skill but allow **greater flexibility and automation**.
- Early Scripting Approaches:**
 - Initially, GIS users had to learn **product-specific languages**:
 - AML (Arc Macro Language)** – used in early ESRI software.
 - Avenue** – used in ArcView 3.x.
- Modern Scripting Languages:**
 - Today, GIS scripts are often written in **industry-standard languages**:
 - Visual Basic for Applications (VBA)**
 - Perl**
 - JScript**
 - Python**

- These languages can:
 - Execute GIS operations.
 - Request **user input**.
 - Display **outputs/results**.
 - Invoke **other software**.

Coupling GIS and Models

- Many GIS systems **lack modeling efficiency**, especially for:
- **Large datasets**
- **Multiple iterations**
- Reason: GIS software is **not primarily designed as a modeling engine**.
- **Workaround:**
- Users often develop **specialized code** (e.g., in **C language**) for faster execution.

Type of Coupling

Description

Example/Notes

Loosely Coupled

Model runs as **separate software**; data exchanged with GIS via files.

Clarke's model.

Often uses **export formats** and **translators** for data exchange (since GIS formats are proprietary).

May require **custom conversion programs** if no common format exists.

Closely Coupled

Both the GIS and the model **read/write the same files**, so **no translation** is needed.

Efficient data exchange.

Embedded

Model is **executed within the GIS** as a **script** or through a **graphic interface** (e.g., ModelBuilder).

Fully integrated execution.

Multi-Criteria Decision Making (MCDM)

- ▶ **MCDM** is a decision-support process that evaluates alternatives by considering multiple quantitative and/or qualitative criteria, assigning weights to them, and combining them to reach a rational and optimal decision.
- ▶ Many real-world decisions depend on combining **multiple weighted factors**.
- ▶ Different stakeholders may value factors differently:
 - ▶ Environmental groups → emphasize ecological impact.
 - ▶ Economists → emphasize cost.
 - ▶ Local communities → emphasize social effects.
- ▶ **Purpose of MCDM:**
 - ▶ To compare and combine such differing viewpoints logically.
- ▶ **Key principle:**
 - ▶ It is better to debate **factor importance and weighting** than to fight over **final decisions**.

Importance of MCDM in Geospatial Analysis

In **GIS-based modelling**, MCDM is widely used to integrate **spatial data layers** (criteria) for decision-making.

It allows planners to:

- ▶ Handle conflicting objectives
- ▶ Incorporate both expert judgment and spatial data
- ▶ Produce suitability or risk maps (e.g., for land use planning, hazard mapping, resource allocation)

GIS and Software Applications

- ▶ Many GIS platforms include MCDM tools.
- ▶ **IDRISI (Clark University)** provides extensive AHP and MCDM capabilities with tutorials.
- ▶ **ArcGIS ModelBuilder** can incorporate weighted overlays and AHP-based modeling.

Model Testing, Validation & Uncertainty

Challenge of Testing Models

- ▶ Models are **complex structures** that often produce **forecasts of the future**.
- ▶ **Testing difficulty:**
 - ▶ Future reality is **not yet available** for comparison.
 - ▶ By the time the future unfolds, interest in testing may have faded.
- ▶ There is a **tendency to trust computer outputs** because:
 - ▶ Results are **numerical** → numbers appear authoritative.
 - ▶ They come from **computers** → seen as objective and precise.
 - ▶ But both can create **false confidence**.

Methods of Model Validation

Historical comparison

- Run the model **forward from a known point in the past**.
- Compare results with **actual historical data**.
- Limitation: same data often used for **calibration**, not independent testing.

Cross-validation

- Split data into two subsets:
 - One for **calibration** (model fitting).
 - One for **validation** (model testing).
- Can be done by dividing data by:
 - **Time** (past vs. future periods).
 - **Space** (different regions).
- Limitation: risky if the system is **non-stationary** (changes over time or space)

Component validation

- Test individual model components to see if they accurately reflect reality.
- Example: Test each rule in Clarke's urban growth model against real land-use changes.
- Limitation: real-world processes rarely behave as simply as model rules.



Thanks

Analytical Hierarchy Process (AHP)

- A widely used MCDM method developed by **Thomas Saaty**.
- Steps:
 1. Identify relevant factors ($x_1, x_2, \dots, x_n$).
 2. Define impact functions ($f(x)$) for each factor.
 3. Ask each stakeholder to **compare factors pairwise** (e.g., slope vs. distance).
 4. Construct a **comparison matrix** for each stakeholder.
 5. Derive **relative weights** (w_i) using matrix analysis (eigenvector method).
 6. Combine all stakeholder matrices to obtain **consensus weights**.
 7. Insert these weights into the GIS model to compute the final **vulnerability index**.

Supervised and Unsupervised Modelling

By

Dr. Meenakshi Devi

EES Department, IISER Mohali

Supervised Learning Models

- Supervised learning is a machine learning approach where models are trained on labeled datasets, meaning each input is paired with a corresponding output. The goal is to learn a mapping function that predicts outputs for new, unseen inputs. This method is widely used in tasks like classification and regression.
- **How Supervised Learning Works**
- The process begins with labeled training data, where the model learns relationships between input features and output labels. The model's parameters are adjusted to minimize prediction errors using optimization techniques like gradient descent. After training, the model is evaluated on a test dataset to measure its performance and ensure it generalizes well to new data.

Common Supervised Modelling Techniques

a) Linear Regression

- Linear regression is used for predictive analysis.
- [Linear regression](#) is a linear approach for modeling the relationship between the criterion or the scalar response and the multiple predictors or explanatory variables.
- Linear regression focuses on the conditional probability distribution of the response given the values of the predictors.
- For linear regression, there is a danger of [overfitting](#). The formula for linear regression is:
- *Syntax:*
- $y = \theta x + b$
- *where,*
- θ - *It is the model weights or parameters*
- b - *It is known as the bias.*
- This is the most basic form of regression analysis and is used to model a linear relationship between a single dependent variable and one or more independent variables.

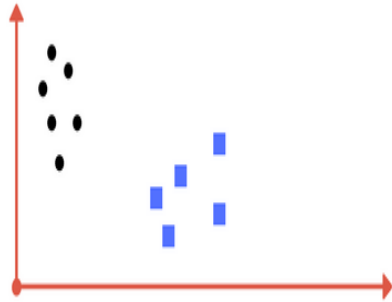
b) Support Vector Machine

- Support vector machines are a set of machine learning techniques based on the concept of **optimal separating hyperplane**, developed by Vapnik (1995).
- In two dimensional space this hyperplane is a line dividing a plane in two parts where in each class lay in either side.
- The hyperplane can be defined as follows:

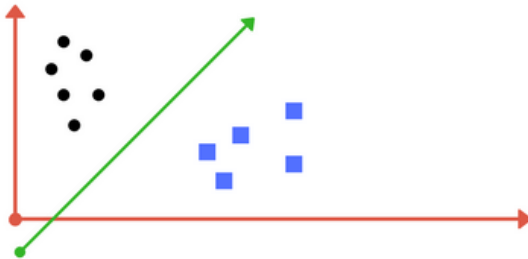
$$f(X) = (W, X) + b = \sum_{i=0}^n (W, X) + b = 0$$

- Where W is the sum of the weights, X is independent variable and b is the normalization constant.

Let's Understand it



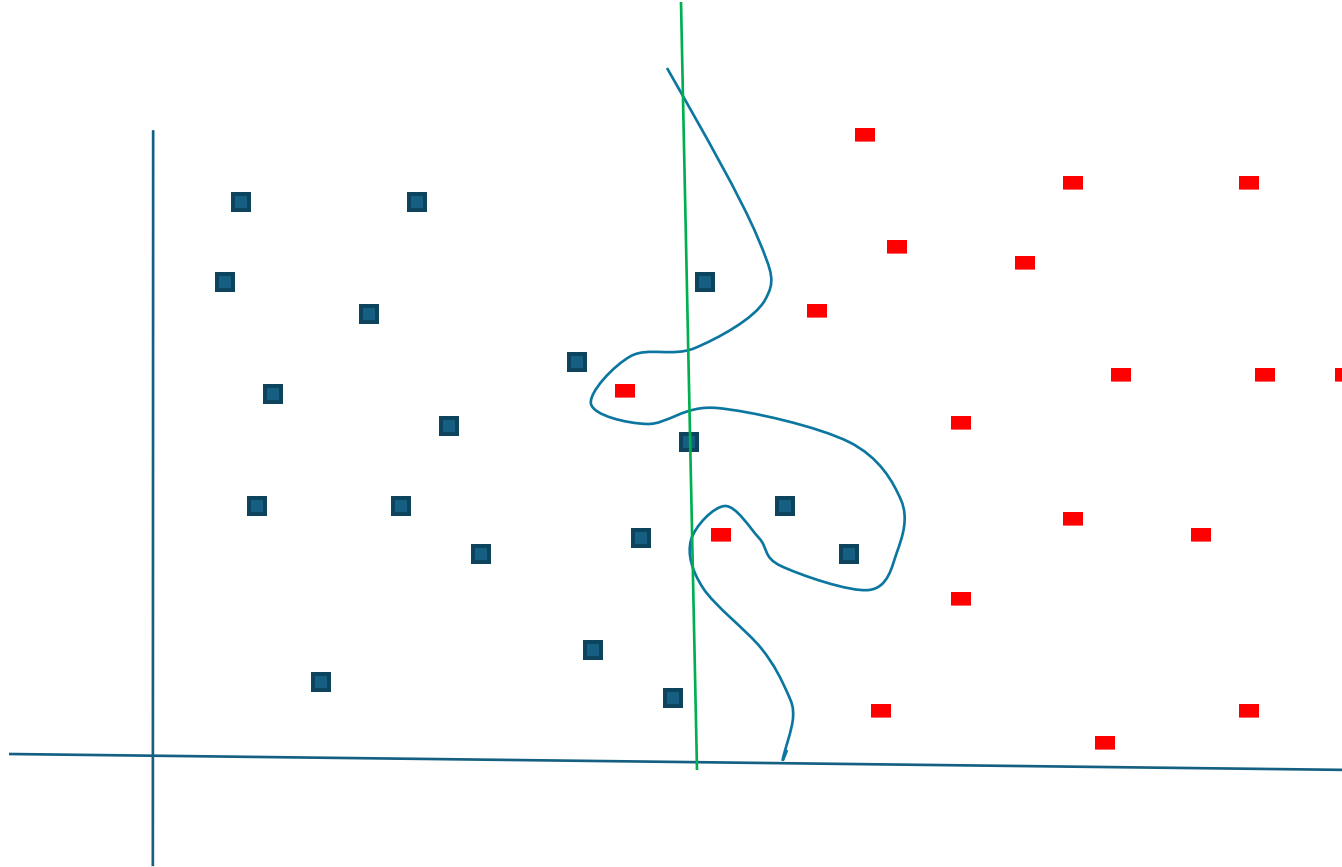
Can you decide a separating line for the classes?



Of course it is very easy



Make it more complicated

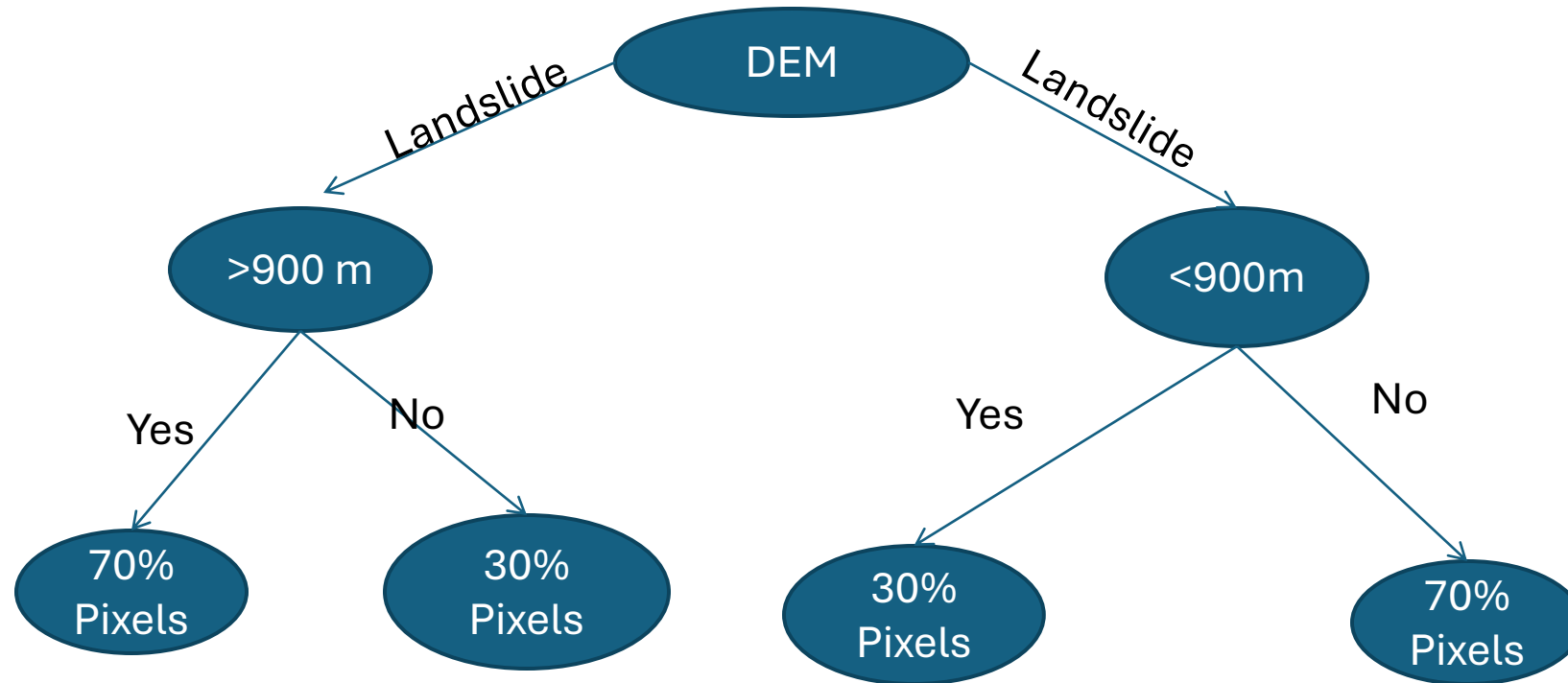


c) Random Forest

Let's First understand Decision Tree.....

- A decision tree is a graph that uses a branching method to illustrate every possible outcome of a decision.
- Decision tree algorithm falls under the category of supervised learning. They can be used to solve both regression and classification problems.

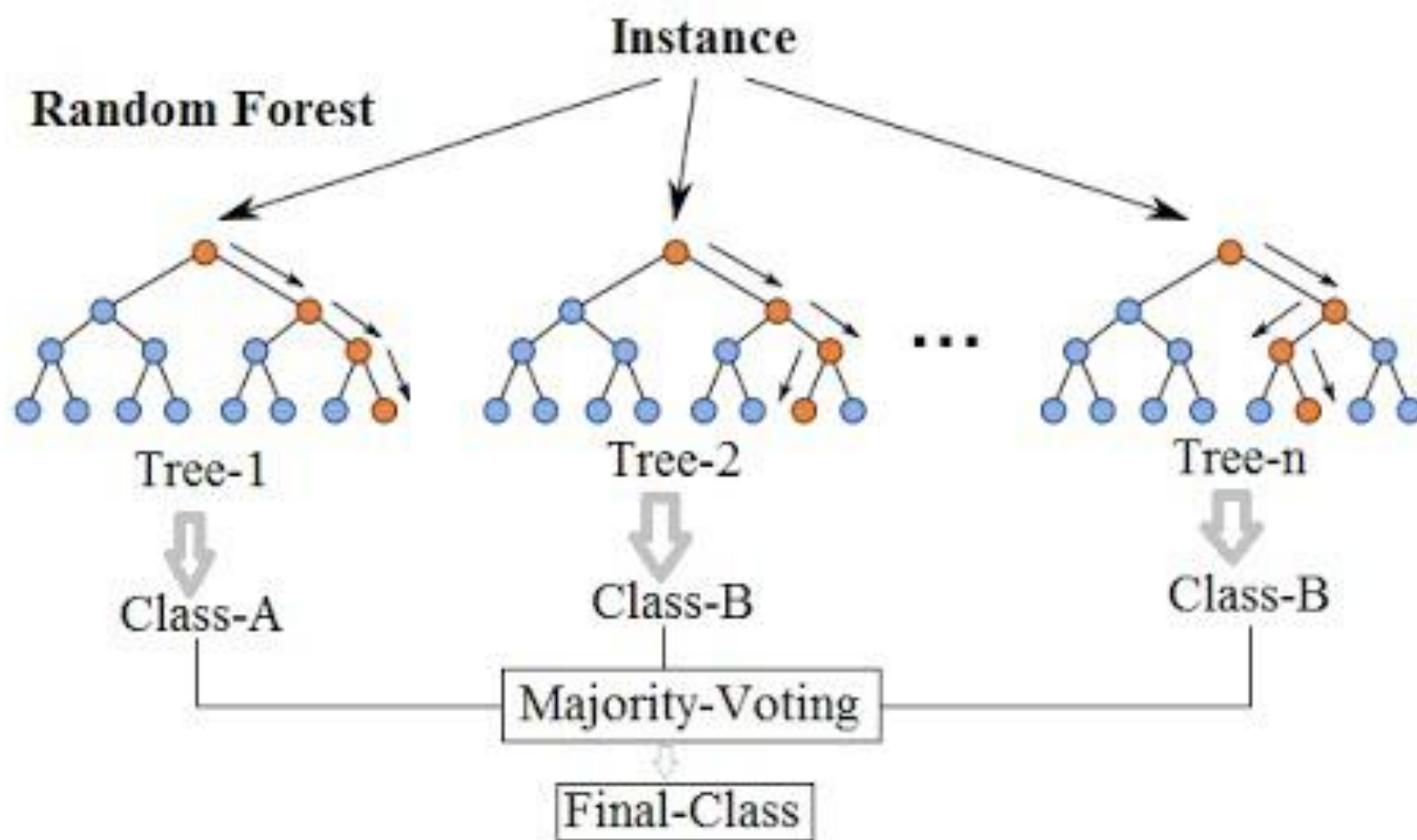
Example



Random Forest

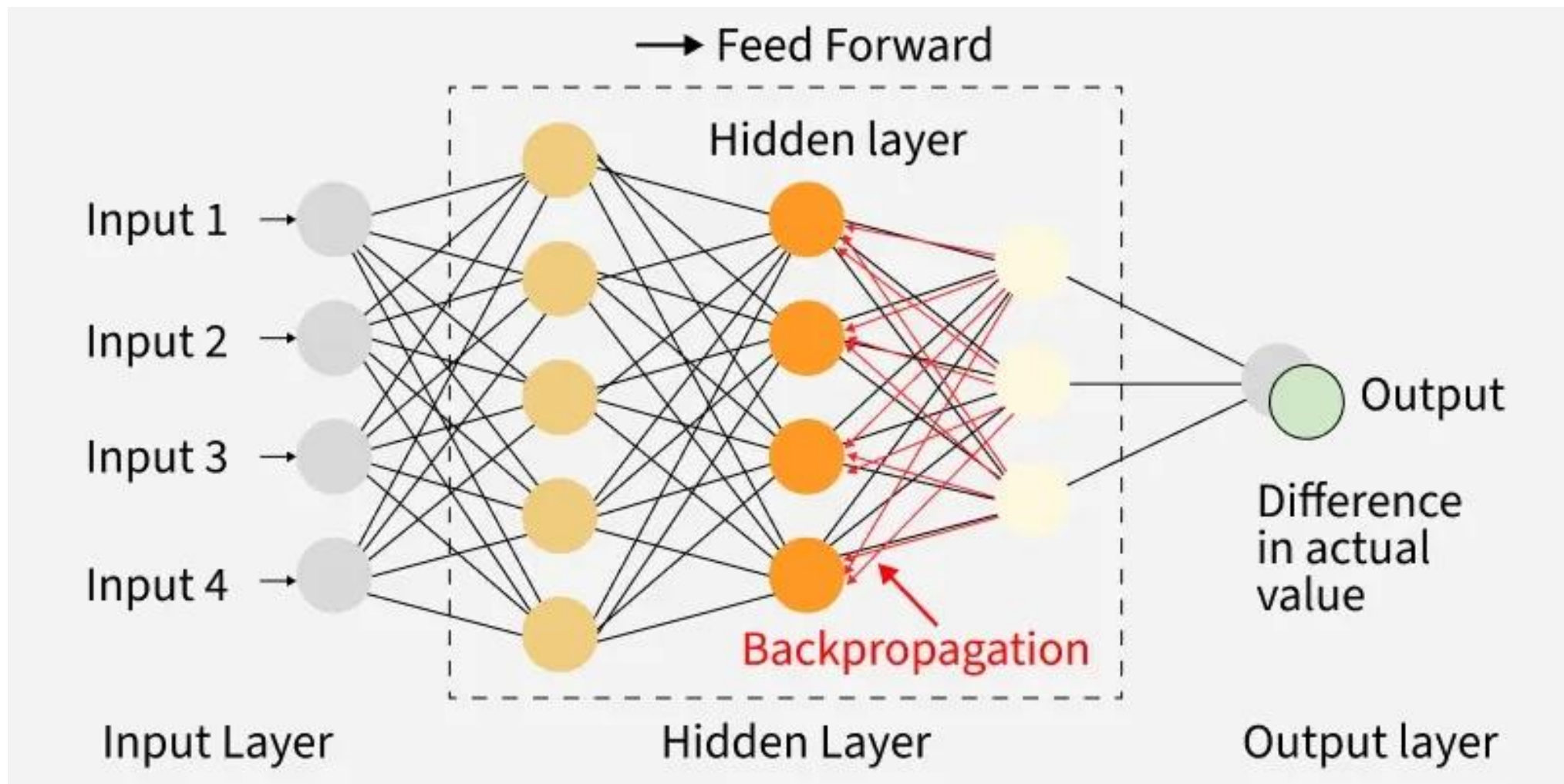
- It is an ensemble machine learning method originally developed by Breiman (2001).
- Less prone to overfitting data problems,
- Unbiased decision making and higher performance.
- In RF, a bootstrap dataset is created by randomly selecting samples from the given dataset.
- The bootstrapping procedure is essentially used to enhance the model performance by decreasing variance without increasing the bias.
- Subsequently, decision trees are built by considering a random subset of variables from selected bootstrap sample and their responses.
- The voted results are ranked and the highest value is selected as the final output of the model.
- During the process of regression and analysis, all the uncorrelated single decision trees are averaged to eliminate errors and noise on individual tree.

Random Forest Simplified



d) Artificial Neural Networks (ANN)

- Artificial Neural Networks (ANNs) are computer systems designed to mimic how the human brain processes information.
- Just like the brain uses neurons to process data and make decisions, ANNs use artificial neurons to analyze data, identify patterns and make predictions.
- These networks consist of layers of interconnected neurons that work together to solve complex problems.
- The key idea is that ANNs can "learn" from the data they process, just as our brain learns from experience.



How do Artificial Neural Networks learn?

- Artificial Neural Networks (ANNs) learn by training on a set of data. For example, to teach an ANN to recognize water body, it will be trained by n no. of water body spectral signature. The network processes these spectral signatures and learns to identify the features that define a water body
- Once the network has been trained, we test it by providing new images to see if it can correctly identify water bodies. The network's prediction is then compared to the actual label . If it makes an incorrect prediction, the network adjusts by fine-tuning the weights of the connections between neurons using a process called backpropagation. This involves correcting the weights based on the difference between the predicted and actual result.
- This process repeats until the network can accurately recognize a water body in an image with minimal error. Essentially, through constant training and feedback, the network becomes better at identifying patterns and making predictions.

Unsupervised Learning Models

- Unsupervised learning is a type of machine learning that works with **unlabeled data**, meaning there are no predefined categories or target variables. The primary goal is to identify **hidden patterns or structures** in the data without any human intervention or prior knowledge.
- **The goal is to identify previously unseen and unconsidered patterns in the data.** Unsupervised learning algorithms analyze data to uncover patterns, relationships, or groupings that are not explicitly labeled or predefined. For example, clustering algorithms can group customers based on purchasing behavior.

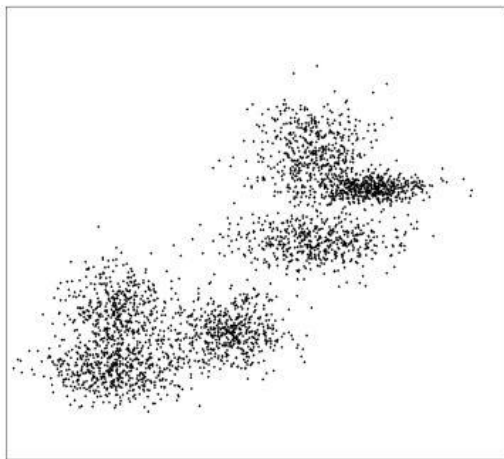
ISO Cluster Analysis

- Clustering is the process of organizing the objects into groups whose members are similar in some way.
- A cluster is therefore a collection of objects which are “similar” between them and are “dissimilar” to the objects belonging to the other clusters.
- Clustering is an unsupervised classification technique.
- It works on the principle of Euclidean metrics and calculate distance between points and center of the cluster.
- The average distance between the clusters can be calculate as follows:

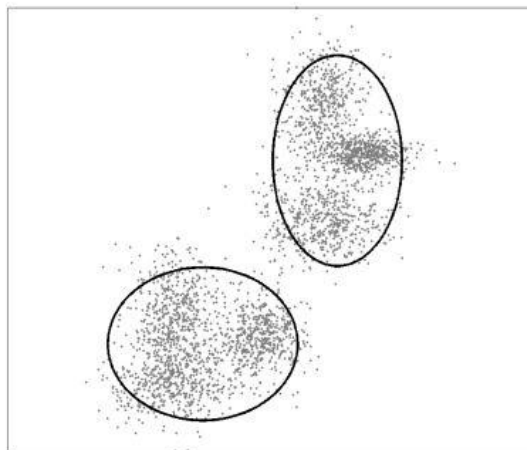
$$AD = 1/N \sum_{i=0}^N \binom{n}{k} AD_i N_i$$

Where AD_i is the average distance of ith pattern in a cluster, N is the total no. of patterns in a given data set.

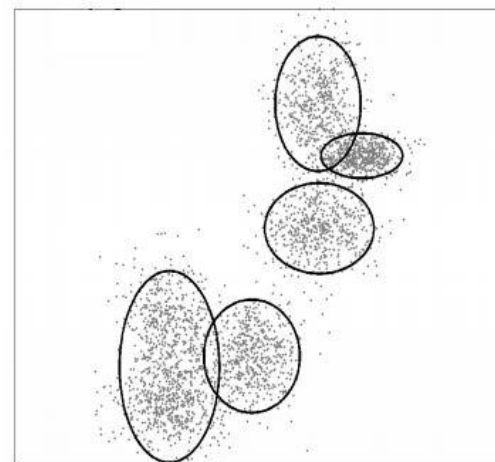
Here is an example



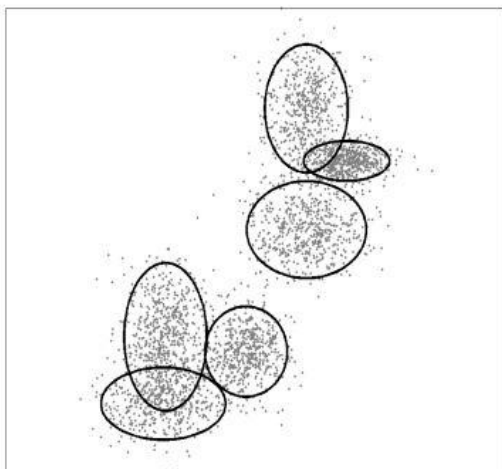
(a) Input data



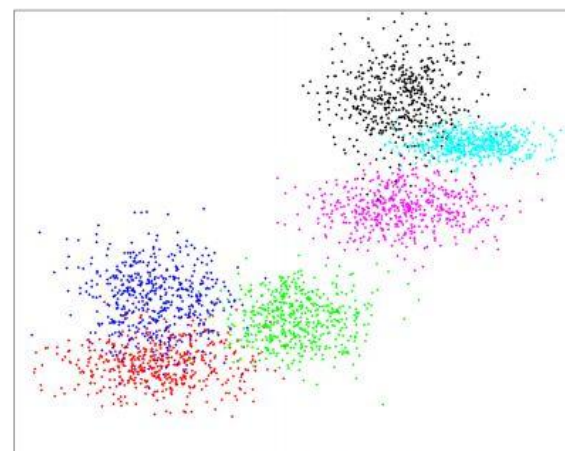
(b) GMM (K=2)



(c) GMM (K=5)



(d) GMM (K=6)



(e) True labels, K = 6

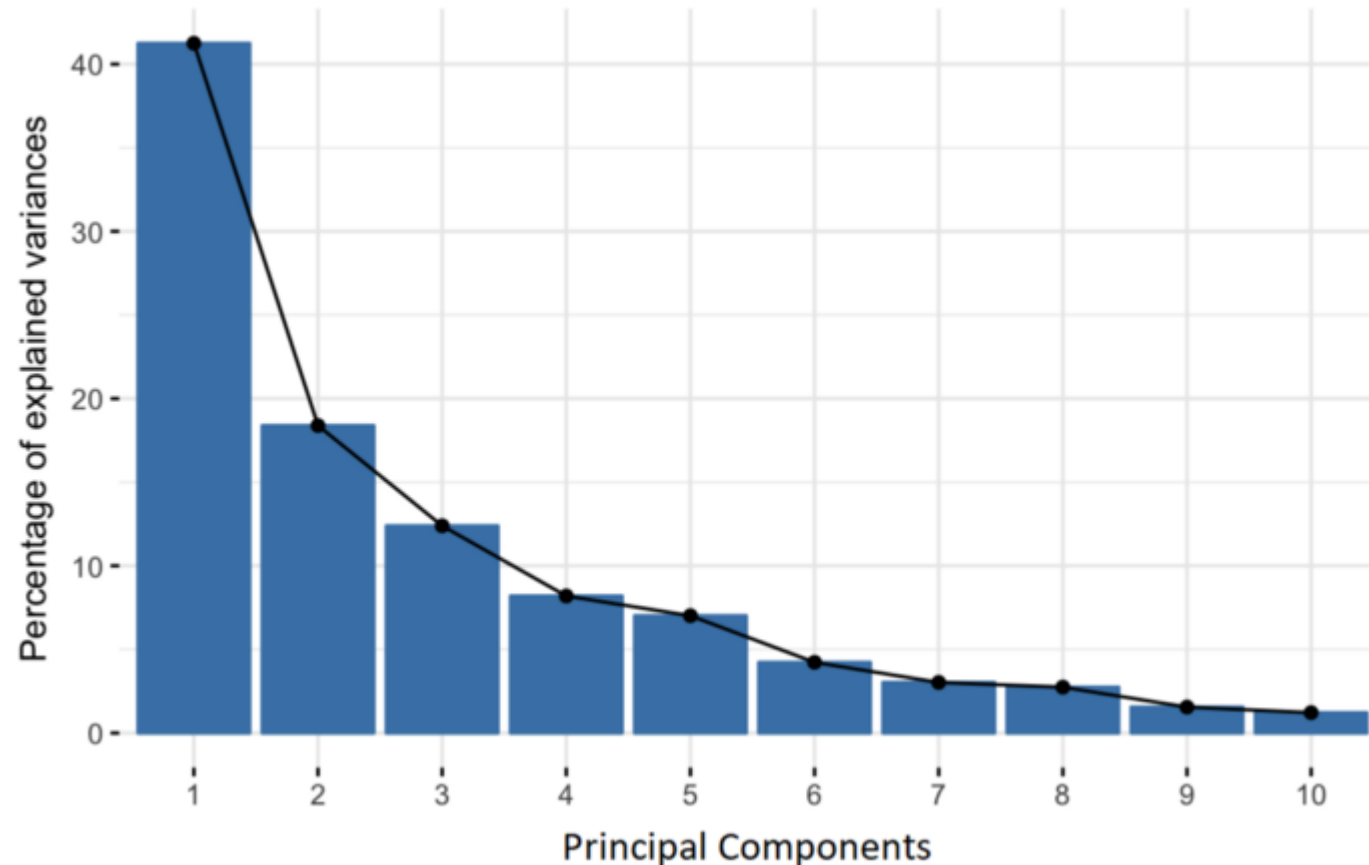
GMM (Gaussian Mixture Model)

Principal Component Analysis

- PCA (Principal Component Analysis) is a dimensionality reduction technique used in data analysis and machine learning. **It helps you to reduce the number of features in a dataset while keeping the most important information.** It changes your original features into new features these new features don't overlap with each other and the first few keep most of the important differences found in the original data.
- PCA is commonly used for data preprocessing for use with machine learning algorithms. It helps to remove redundancy, improve computational efficiency and make data easier to visualize and analyze especially when dealing with high-dimensional data.

What are Principal Components

- **New variables that are constructed as linear combinations or mixtures of the initial variables.** new variables (i.e., principal components) are uncorrelated and most of the information within the initial variables is squeezed or compressed into the first components.



So, the idea is 10-dimensional data gives you 10 principal components, but PCA tries to put maximum possible information in the first component, then maximum remaining information in the second and so on, until having something like shown in the scree plot .

Model evaluation and validation

- To quantitatively assess the performance and validation of prediction model, statistical metrics such as sensitivity, specificity, area under curve (AUC), accuracy and Kappa index, are generally used.
- All these parameters are represented by the following equations:

$$Sensitivity = \frac{TP}{(TP + FN)}$$

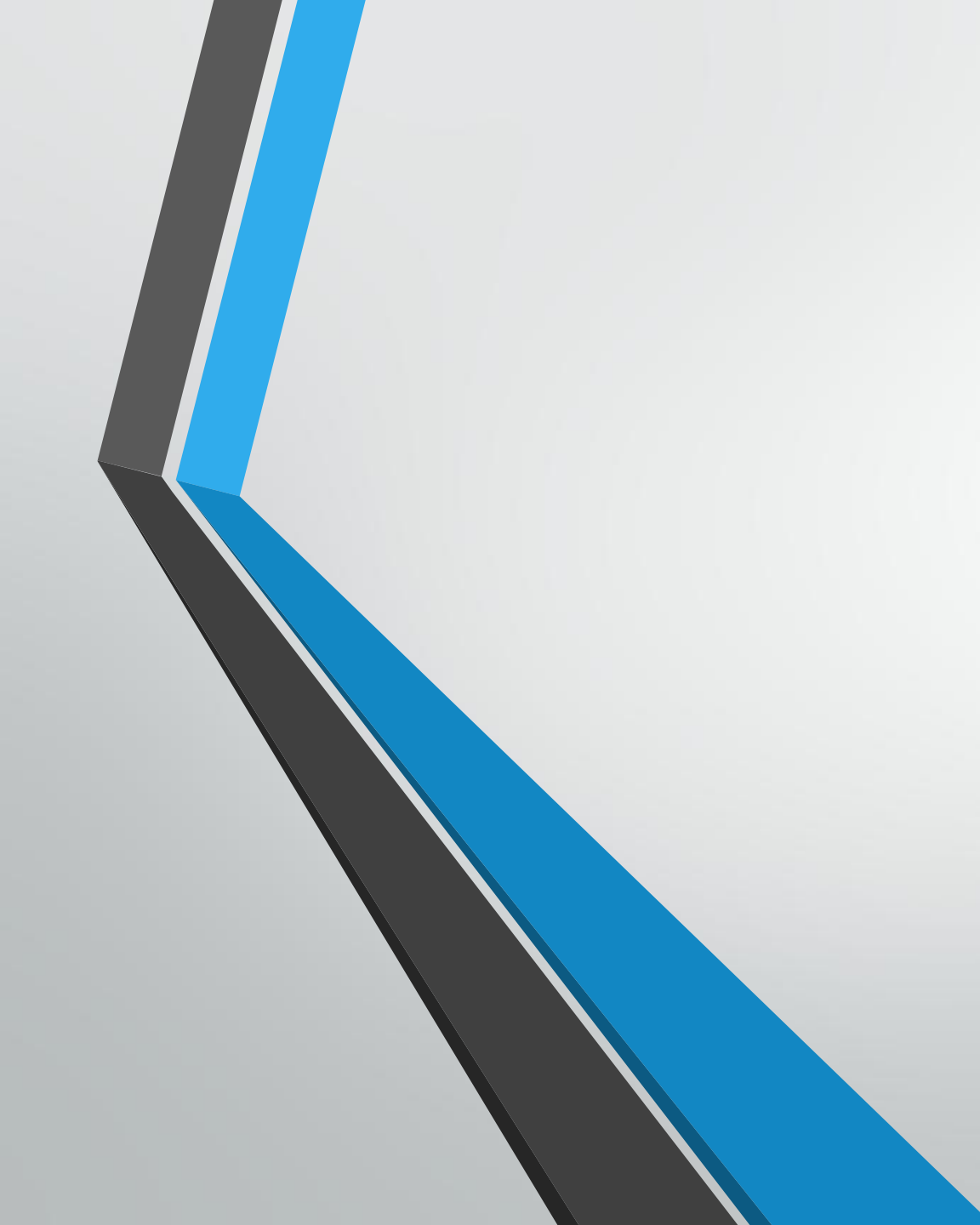
$$Specificity = \frac{TN}{(TN + FP)}$$

$$AUC = \left(\frac{\sum TP + \sum TN}{P + N} \right)$$

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

$$Kappa = \frac{P_o - P_e}{1 - P_e}$$

Where TP is the true positive values, TN is the true negative values. FP and FN represents false positive and false negative, respectively. N is the total no. of values in training data, and P_o and P_e is the observed and predicted values .



Data Fusion

By

Dr. Meenakshi Devi

EES Department, IISER Mohali

Definition of Data Fusion:

Data Fusion is the formal framework and set of tools for the alliance of data originating from different sources. It aims to **produce information of greater quality and reliability** that could be achieved by using a single source alone.

- The integration of remotely sensed data with GIS data to create a unified, information-rich dataset that supports enhanced analysis, modeling, and decision-making.

Levels of Data Fusion

Pixel-Level (or Low-Level) Fusion

- Combination of raw data from multiple sensors or sources at the most basic data element level—the pixel.
- To create a new, fused image with improved information content.
- **Examples:**
 - **Pan-sharpening:** Fusing a high-spatial-resolution panchromatic image (B&W) with a lower-spatial-resolution multispectral image (color) to create a high-resolution color image.
 - **Hyperspectral & Multispectral Fusion:** Merging datasets to achieve both high spectral and spatial detail.
- **Advantages:** Preserves original data as much as possible.
- **Disadvantages:** Computationally intensive; requires precise geometric and radiometric registration.

Feature-Level Fusion

Extraction of salient features (e.g., edges, textures, shapes, lines) from each source first, followed by the fusion of these feature sets.

- **Goal:** To create a more robust and complete set of features for analysis or classification.
- **Examples:**
 - Extracting roads from a high-resolution image and fusing them with road centerlines from a GIS vector database to update the database.
 - Using texture from a SAR (Synthetic Aperture Radar) image and spectral signatures from an optical image to classify urban structures.
- **Advantages:** More compact representation, reduces data volume.
- **Disadvantages:** Dependent on the accuracy of the initial feature extraction.

Decision-Level (or High-Level) Fusion

- Each data source is processed independently to yield a separate decision or classification map. These individual results are then combined using logical or statistical methods to reach a final, consensus decision.
- **Goal:** To achieve a more reliable and accurate final product than any single classifier could produce.
- **Examples:**
 - **Majority Voting:** Taking a land classification from an optical image and one from a LiDAR-derived model, and for each pixel, choosing the class that the majority of sources agree on.
 - **Bayesian Methods:** Using probability theory to combine the results.
 - **Fuzzy Logic:** Handling uncertainty in class membership.
- **Advantages:** Most flexible, can fuse disparate data types (e.g., image + expert knowledge); tolerant to failures in one sensor.
- **Disadvantages:** Loss of original sensor information; fusion logic can be complex.

The prerequisites of pixel based fusion

The actual procedure of fusion does not consume much time and energy. It is the preprocessing work and the efforts to bring the diverse data to a common platform that requires a greater attention and effort.

Proper Pre-Processing of the image data

- ✓ Fusion involves the interaction of the images having different spatial resolution.
- ✓ These images have different pixel size, which creates problems while merging.
- ✓ Hence the image data is resample to a common pixel spacing and map projection.
- ✓ The common pixel spacing should be the one, which is required in the desired fused image.
- ✓ After the fusion process the contribution of each sensor cannot be distinguished or quantified. Therefore as a general rule one must attempt to produce the best single sensor geometry and radiometry and then fuse the images. Any spatial enhancement performed prior to image fusion is of benefit to the resulting fused image.



Techniques of Pixel based fusion

Multiplicative Technique

- The Multiplicative equation is given by Crippen, Robert E. 1989.
- The algorithm shown below operates on the original image. The result is an increased presence of the **intensity component**. For many applications, this is desirable.
- People involved in urban or suburban studies, city planning, and utilities routing often want roads and cultural features (which tend toward high reflection) to be pronounced in the image.

Red = (Low Resolution Band1 * High Resolution Band1)

Green = (Low Resolution Band2 * High Resolution Band2)

Blue = (Low Resolution Band3 * High Resolution Band3)



Merged PAN+ Multispectral using Multiplicative tech.

BROVEY TRANSFORM

Brovey transform is a formula that normalizes multispectral bands used for a RGB display, and multiplies the result by any desired data to add the intensity or the brightness component of the image.

The Brovey transform equation is given by

$$DNB_{1new} = \frac{DN B_1}{DN B_1 + DNB_2 + DNB_3} \times DN \text{ high Resolution Image}$$

$$DNB_{2new} = \frac{DN B_2}{DN B_1 + DNB_2 + DNB_3} \times DN \text{ high Resolution Image}$$

$$DNB_{3new} = \frac{DN B_3}{DN B_1 + DNB_2 + DNB_3} \times DN \text{ high Resolution Image}$$

BROVEY TRANSFORM contd..

- The Brovey Transform was developed to visually increase contrast in the low and high ends of an image's histogram (i.e., to provide contrast in shadows, water and high reflectance areas such as urban features).
- Consequently, the Brovey Transform should not be used if preserving the original scene radiometry is important.
- However, it is good for producing RGB images with a higher degree of contrast in the low and high ends of the image histogram and for producing visually appealing images.
- Since the Brovey Transform is intended to produce RGB images, only three bands at a time should be merged from the input multispectral scene, The resulting merged image should then be displayed with bands 1, 2, 3 to RGB.

Image Fusion using IHS transformation

(intensity Hue Saturation)

Procedure

1 RGB to IHS transformation

Three bands of low spatial resolution data in RGB are transformed into three I,H,S components

2. Contrast Enhancement

The high spatial resolution image (Panchromatic) is contrast stretched so that it has approximately the same variance and mean as intensity image

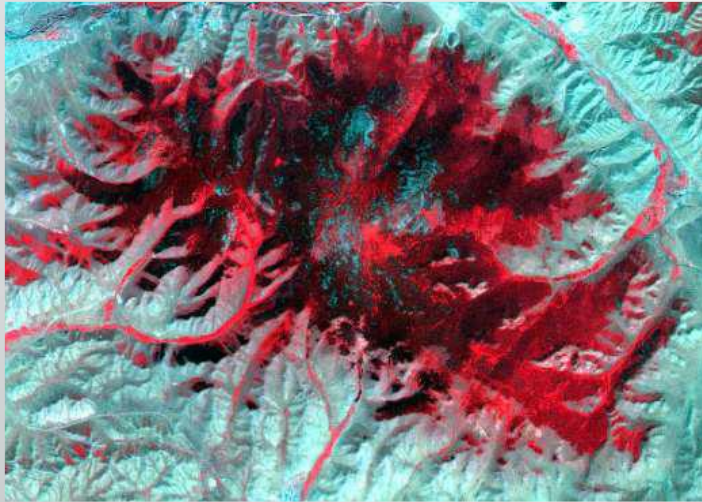
3. Substitution

The stretched high spatial resolution image is substituted for the intensity (I) Image.

4. IHS to RGB

The modified IHS dataset is transformed back to RGB color space using an inverse IHS transformation.

Image Fusion using IHS transformation



PRINCIPAL COMPONENT ANALYSIS

- The multispectral image data is usually strongly correlated from one and to the other thus contain similar information.
- For example, Landsat MSS Bands 4 and 5 (green and red, respectively) typically have similar visual appearances since reflectances for the same surface cover types are almost equal
- Principal component analysis is a pre-processing transformation that creates new images of with uncorrelated components as band.
- The objective of this transformation is to reduce the dimensionality (i.e. the number of bands) in the data, and compress as much of the information in the original bands into fewer bands.
- The "new" bands that result from this statistical procedure are called components.
- Principal component images may be analysed as separate black and white images, or any three component images may be colour coded to form a colour composite.

Image Fusion using PCA

Procedure

- ✓ Principal component images are derived from raw image data set
- ✓ Panchromatic data is stretched to have approximately the same variance and average as the first principal component
- ✓ The stretched PAN image is substituted for first principal component and the data are transformed back into RGB space

(the reason for substituting PAN image for PC₁ is that normally PC₁ contains all the information unique to all the bands input to PCA)

Advanced Integrated Fusion Techniques

- **Statistical/Machine Learning Methods:**
 - **Regression Techniques:** Modeling the relationship between coarse and fine-resolution data to "downscale" information (e.g., for soil moisture).
 - **Deep Learning (CNN):** Using Convolutional Neural Networks to learn complex, non-linear relationships between different data sources for tasks like pan-sharpening, classification, and feature extraction. This is the current state-of-the-art.

Practical Applications

- **Precision Agriculture:** Fusing satellite multispectral data (crop health) with GIS data on soil conductivity and yield history to create variable rate application maps for fertilizers.
- **Disaster Management:** Fusing pre-event satellite imagery, post-event SAR data (which can see through clouds), and GIS layers of infrastructure and population density to assess damage and plan response.
- **Urban Planning:** Fusing high-resolution imagery, LiDAR data (for 3D building models), and GIS zoning maps to monitor urban sprawl, assess solar potential, or plan new infrastructure.
- **Environmental Monitoring:** Fusing time-series of satellite imagery (e.g., for deforestation) with GIS protected area boundaries and topographic data to model and combat illegal logging.
- **Hydrology:** Fusing radar altimetry data (water level) with GIS-based digital elevation models to model flood inundation extents.



Applications of GIS

BY

DR. MEENAKSHI DEVI

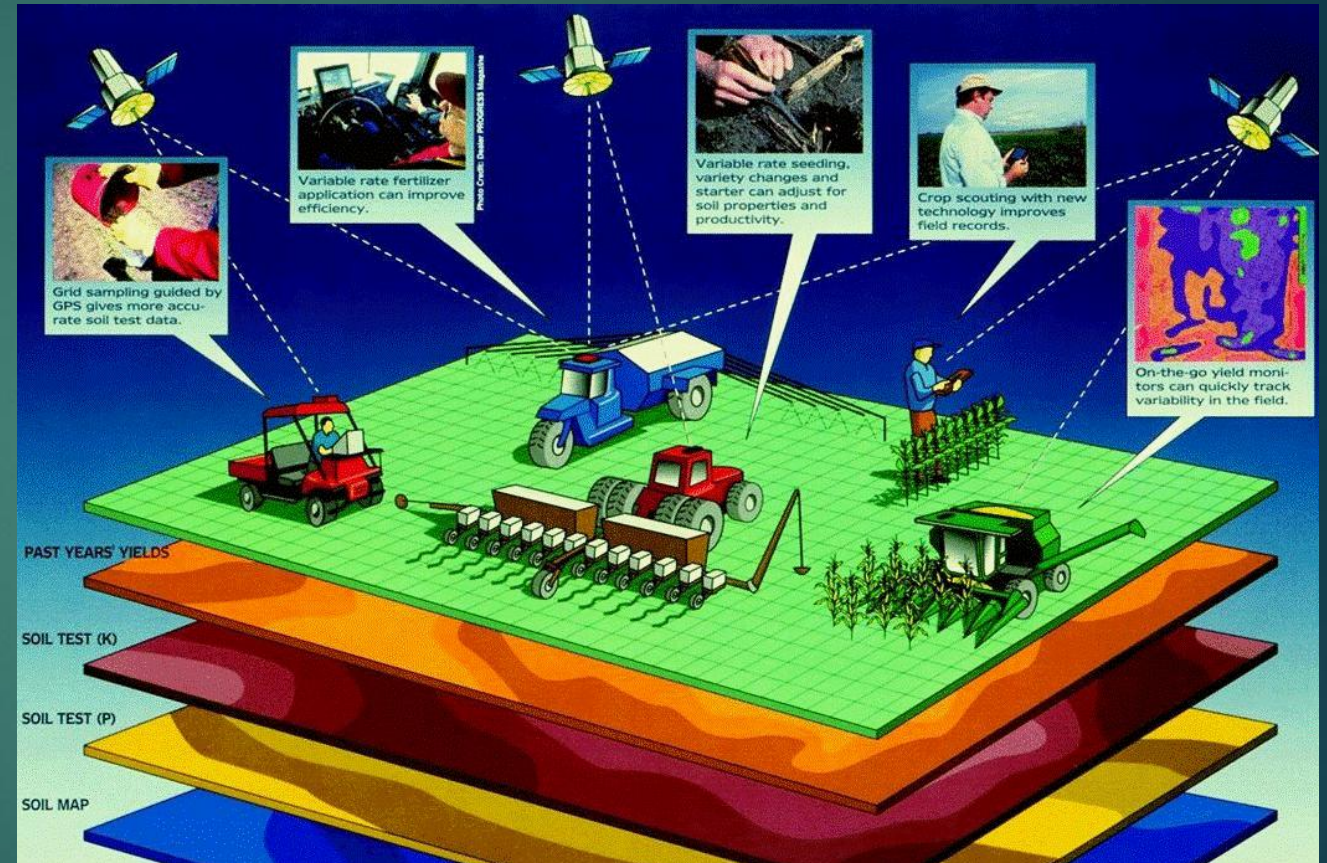
EES DEPARTMENT, IISER MOHALI

Introduction

- ▶ *GIS (Geographic Information System) is a technology that collects data to create a special digital map. This map has different layers that show where all living things (humans, water sources, trees) and non-living things (buildings, roads, vehicles) are situated.*
- ▶ It integrates spatial (location-based) and non-spatial (attribute) data to support decision-making and problem-solving in diverse fields.
- ▶ The real strength of GIS lies in its ability to analyze spatial relationships and patterns, offering insights that traditional data analysis cannot provide.
- ▶ Geographic Information Systems can be helpful in various domains. Some of the most common among them are:

1. Agriculture

- ▶ GIS is a useful tool in agriculture. It uses data about farms, weather, and water resources to help determine the best agricultural practices. Researchers can assist farmers in finding suitable weather conditions for specific crops and locating nearby water sources for irrigation. Governments can utilize GIS to create resource management policies and identify appropriate agricultural lands.



Applications of GIS in Agriculture:

- ▶ **Drought management:** GIS helps manage droughts by identifying areas or lands that are experiencing a lack of water. It helps prevent damage to plant seeds, human efforts, and expensive fertilizers.
- ▶ **Pest control:** It is also used for pest control by predicting attacks from pests like locusts and rodents by analyzing spatial data of specific agricultural lands. It allows the government and farmers to plan how to deal with these pests.
- ▶ **Land and soil analysis:** By studying previous data sets, satellite images, or even analyzing the field directly, field workers can gather information about the land and soil conditions.
- ▶ **Planning of future food demand:** By considering the population's needs, the government and farmers can plan and produce enough crops to meet the demand.

2. Environment

GIS can monitor how clean the air is in a specific area. It uses data from satellites that monitor air quality to create maps that show places with high pollution. Government can use this map to create policies to improve the air quality in those places.



Applications of GIS in Environment:

- ▶ Monitoring land-use and land-cover (LULC) changes through remote sensing integration.
- ▶ Mapping deforestation, desertification, and biodiversity loss.
- ▶ Identifying pollution sources and assessing air and water quality.
- ▶ Modeling soil erosion, flood risk, and watershed management.
- ▶ Example: Use of GIS in Environmental Impact Assessment (EIA) to identify vulnerable areas and predict impacts.
- ▶ Monitoring the air and water quality to analyze real-life habitat conditions.
- ▶ Selection of proper location for new infrastructure plans by checking the project's impact on the environment.
- ▶ **Resource management:** Governments can locate areas with natural resources like water bodies, forests, and agricultural lands.
- ▶ **Climate change:** We can analyze climate data to assess the risks associated with climate change.

3. Urban Planning and Transportation

GIS helps city planners and transportation experts combine different types of information like maps, satellite pictures, population statistics, and infrastructure data. It helps them make better decisions when designing cities and transportation systems that are sustainable and good for the environment.



Applications of GIS in Urban Planning and Transportation:

- ▶ **Growth Forecast:** It's useful in developing models to predict and plan for the future expansion of cities.
- ▶ **Planning Land Usage:** It lets infrastructure experts analyze and allocate land for various purposes like residential, commercial, and industrial use in urban areas.
- ▶ **Transportation network:** Countries can improve transportation systems, including roads, public transit, and pedestrian infrastructure, to ensure smooth movement within cities.
- ▶ **Infrastructure management:** Cities can efficiently manage urban infrastructure, such as bridges and public facilities, to support sustainable development.

4. Disaster Management

GIS can keep an eye on places that have a higher chance of natural disasters. So, it can alert the environment-related authorities in case of emergencies, and they can act accordingly. It can help them reach the affected areas in time, respond to victims, develop recovery plans, and take measures to prevent future disasters.

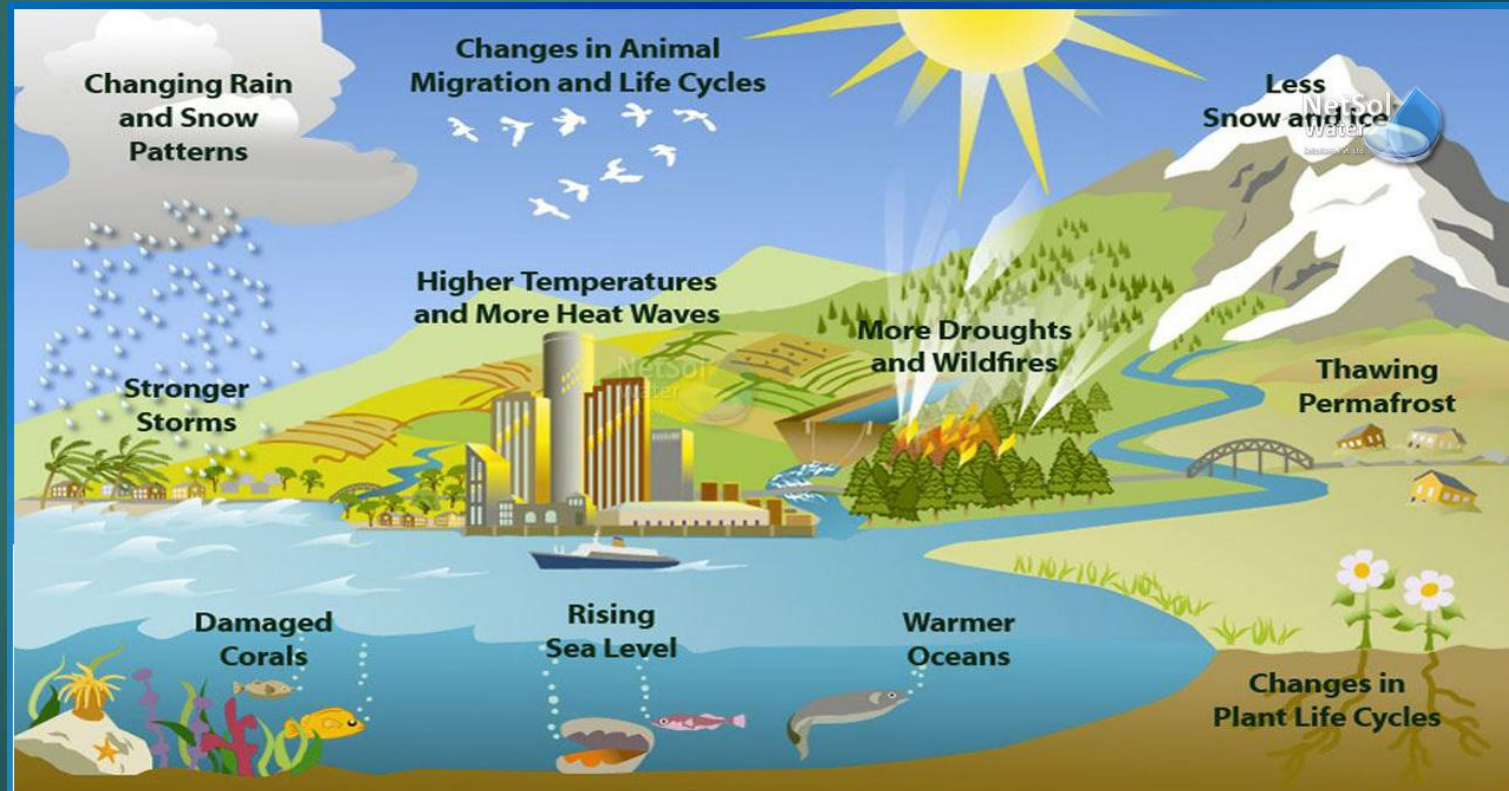


Applications of GIS in Disaster Management:

- **Hazard Zonation Mapping:** Flood, landslide, earthquake, and cyclone risk areas.
- **Vulnerability and Risk Assessment:** Integration of demographic and infrastructural data to assess exposure.
- **Disaster Response:** Real-time mapping for relief distribution and damage assessment.
- **Recovery Planning:** Identifying safe zones and reconstruction priorities.
- **Example:** GIS-based **Flood hazard Modelling** using DEM and hydrological models.

5. Application of GIS in Natural Resource Management

Natural Resource Management (NRM) involves the sustainable utilization, monitoring, and conservation of resources such as land, water, soil, minerals, and forests. GIS (Geographic Information System) plays a vital role in collecting, analyzing, and visualizing spatial data for effective resource management and planning. It helps integrate various thematic datasets (e.g., geology, vegetation, topography) to support decision-making and policy formulation.



Management and Conservation of Natural Resources

Primary applications

a) Land Resource Management

- **Land Use/Land Cover (LULC) Mapping:**

- Integration of satellite imagery and GIS helps in identifying different land-use categories (agriculture, forest, settlement, wasteland).
- Enables **change detection analysis** to study urban expansion, deforestation, or degradation.

- **Soil Erosion and Degradation Mapping:**

- GIS-based models (e.g., USLE, RUSLE) assess erosion-prone zones.
- Helps in planning soil conservation measures.

- **Land Capability Classification:**

- Evaluates land suitability for agriculture, forestry, or construction.

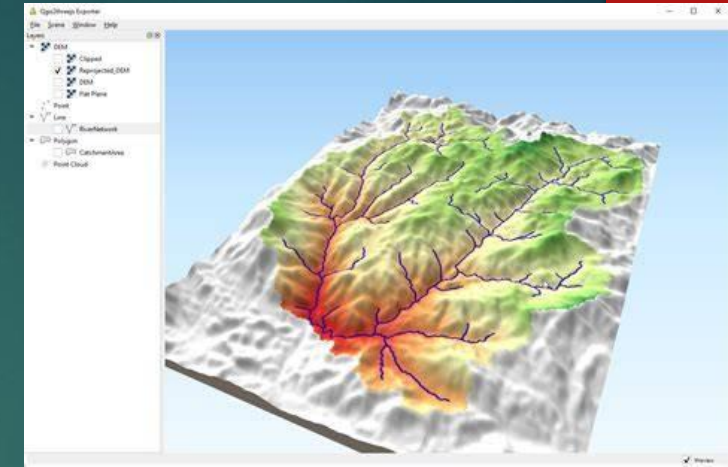
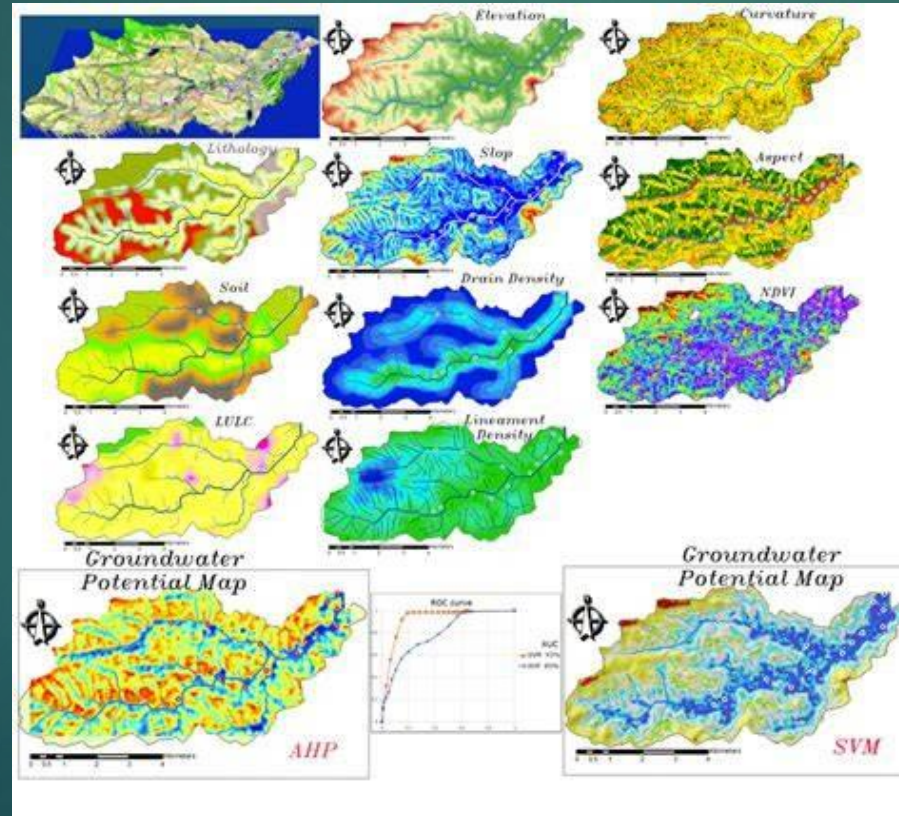
- **Wasteland Mapping:**

- Identifies degraded lands for reclamation or afforestation.

b) Water Resource Management

- **Watershed Delineation and Management:**

- Using DEM (Digital Elevation Model), GIS delineates watersheds and sub-watersheds.
- Prioritizes micro-watersheds based on morphometric and hydrological parameters.

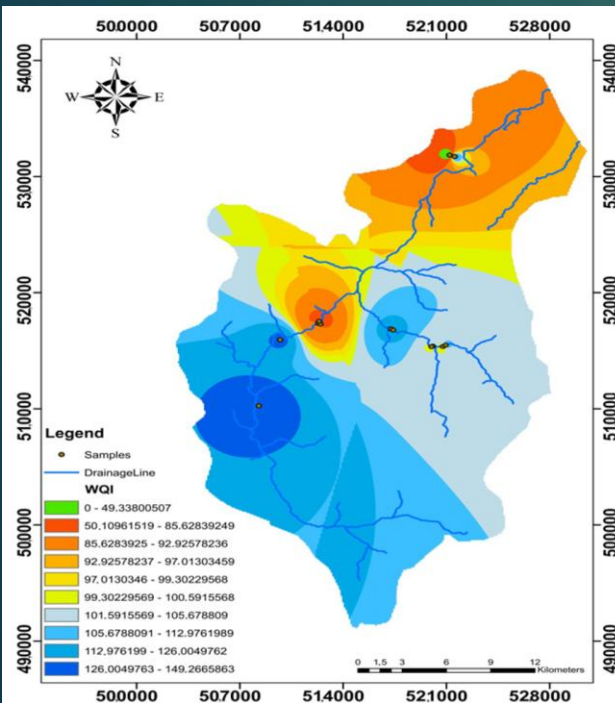


- **Groundwater Potential Mapping:**

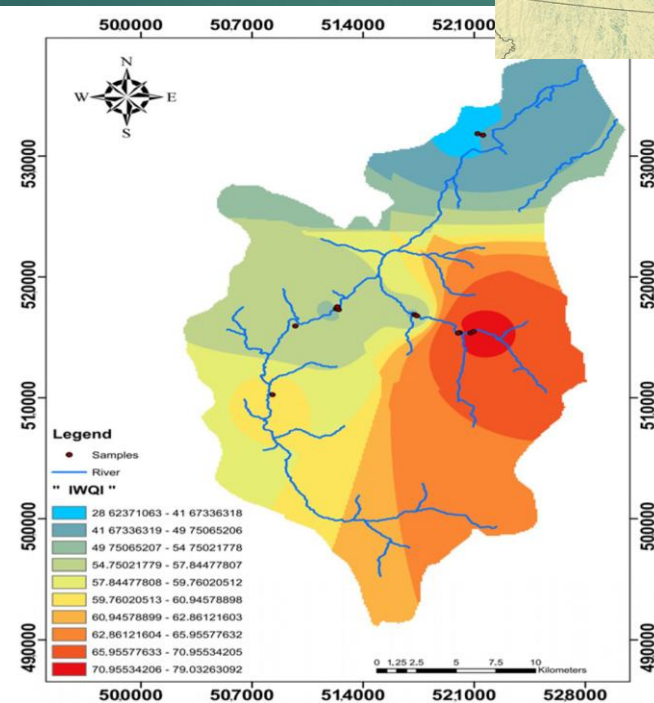
- Integrates geology, slope, drainage density, land use, and rainfall data to locate recharge zones.

- Surface Water Analysis:**

- Mapping of lakes, rivers, and reservoirs for planning irrigation or hydropower projects.



(a)



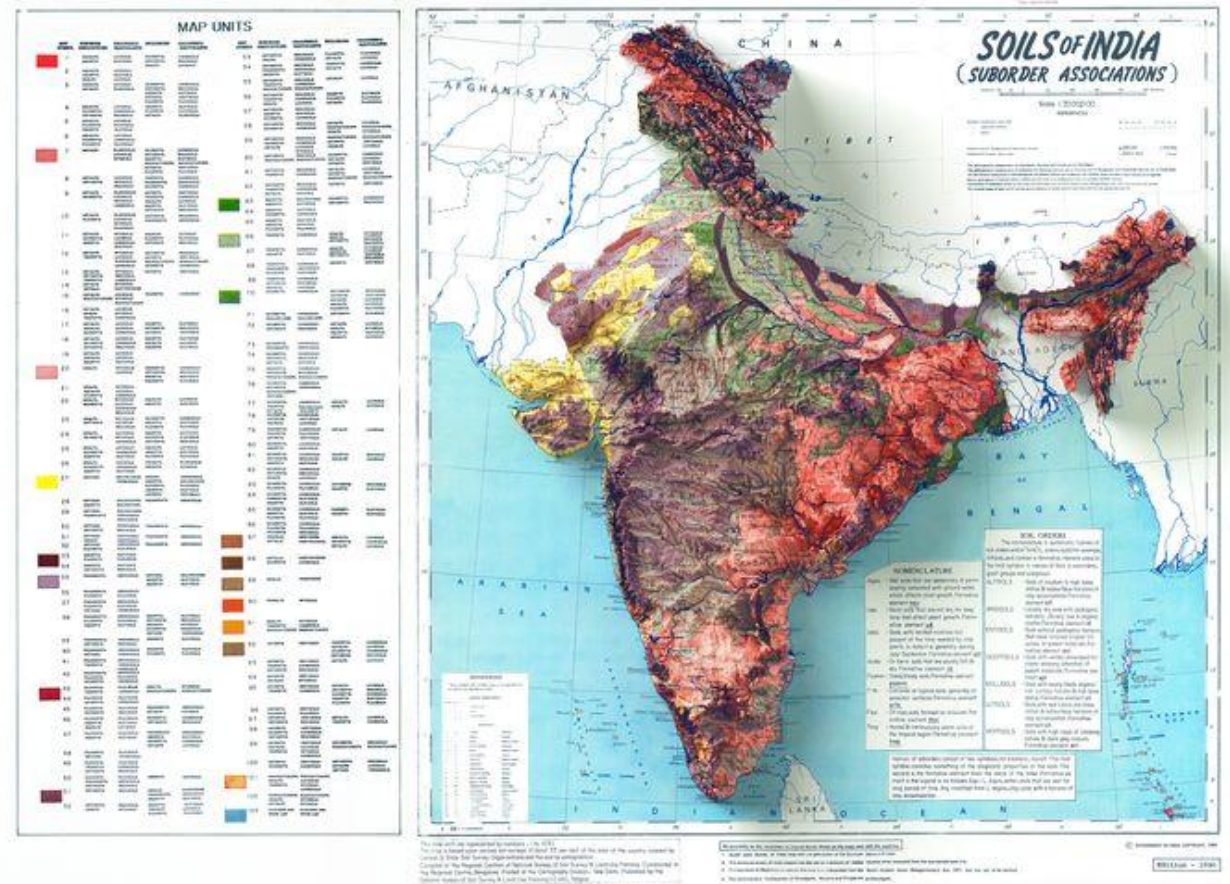
(b)

- Water Quality Assessment:**

- Spatial interpolation (e.g., Kriging) of water quality parameters (pH, TDS, nitrate) for monitoring pollution.

c) Soil Resource Management

- **Soil Mapping and Classification:**
- GIS helps in generating digital soil maps showing texture, pH, nutrient status, and moisture content.
- **Soil Fertility Assessment:**
- Spatial interpolation of soil nutrient data to guide fertilizer management.
- **Erosion Risk Mapping:**
- Combines rainfall, slope, and land cover data to estimate erosion susceptibility.
- **Soil Moisture Modelling:**
- Used for drought assessment and irrigation planning.



d). Forestry

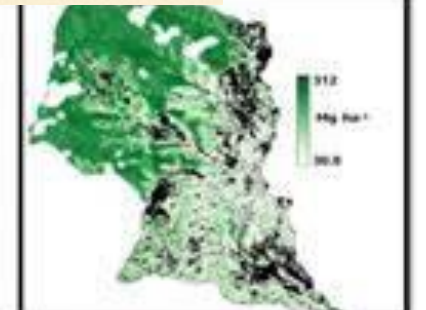
- ▶ GIS and remote sensing data are used to classify forest cover and canopy density.
- ▶ Regular monitoring supports the National Forest Inventory and Forest Survey of India (FSI) reports.
- ▶ Time-series satellite data detect forest loss and regeneration trends.
- ▶ Identifies habitats, migration corridors, and fragmentation zones for conservation planning.
- ▶ GIS integrates slope, vegetation, and climatic data to map fire-prone regions.
- ▶ Estimation of biomass and carbon stock for climate change mitigation.



(a) Agroforestry mapping
Platform: Satellite, Airborne (UAV)
Dataset: MS, MS+SAR, MS+HS, MS+LiDAR, LiDAR + HS



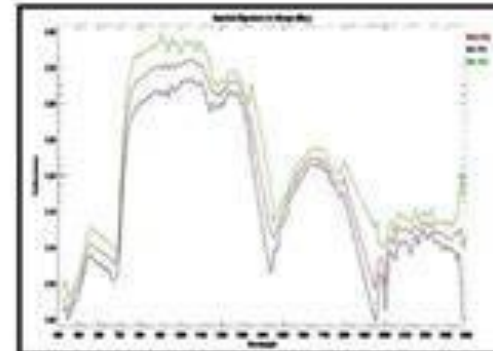
(b) Agroforestry Suitability mapping
Platform: Satellite
Dataset: Radar



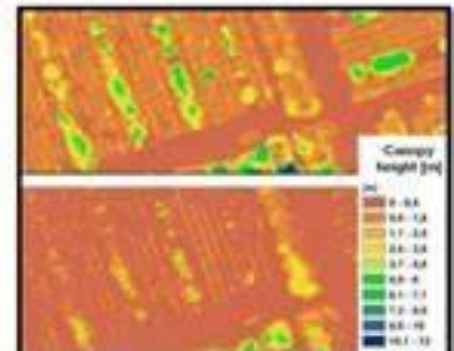
(c) Biomass & CS
Platform: Satellite, Airborne (UAV)
Dataset: Radar, models, equations



(d) Canopy layering
Platform: Airborne (UAV)
Dataset: LiDAR, MS+LiDAR, LiDAR + HS



(e) Agroforestry tree signature
Platform: Satellite, Airborne (UAV)
Dataset: HS



(f) Canopy height
Platform: Airborne (UAV)
Dataset: MS, MS+LiDAR